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Analyzing Uncertainty Quantification in Statistical and Deep Learning Models for Probabilistic Electricity Price Forecasting

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ABSTRACT Precise probabilistic forecasts are fundamental for energy risk management, and there is a wide range of both statistical and machine learning models for this purpose. Inherent to these probabilistic models is some form of uncertainty quantification. However, most models do not capture the full extent of uncertainty, which arises not only from the data itself but also from model and distributional choices. In this study, we examine uncertainty quantification in state-of-the-art statistical and deep learning probabilistic forecasting models for electricity price forecasting in the German market. In particular, we consider distributional deep neural networks (DDNNs) and augment them with an ensemble approach, Monte Carlo (MC) dropout, LASSO-estimated autoregressive (LEAR), and conformal prediction to account for model uncertainty. Additionally, we consider the eXtreme Gradient Boosting (XGBoost) approach combined with quantile regression averaging (QRA), generalized autoregressive conditional heteroskedasticity (GARCH), and conformal prediction. Across a range of performance metrics, we find that the XGBoost-based models perform well in terms of probabilistic forecasting, irrespective of the uncertainty quantification method. Furthermore, we find that DDNNs benefit from incorporating both data and model uncertainty, improving both point and probabilistic forecasting. Uncertainty itself appears to be best captured by the models using conformal prediction. Overall, our extensive study shows that all models under consideration perform competitively. However, their relative performance depends on the choice of metrics for point and probabilistic forecasting.

INDEX TERMS Bayesian methods, deep learning, electricity price forecasting, ensemble learning, interval forecasting, machine learning, Monte Carlo methods, power markets, probabilistic forecasting, uncertainty quantification

I. INTRODUCTION

THE energy industry has been in turmoil lately. On the positive side, we see a considerable increase in renewable energy, leading to a reduced carbon footprint of power production. However, this comes with a drawback, as renewable power can be highly fluctuating, which leads to increased production risk and larger price volatility in turn. This volatility is further amplified by various geopolitical developments in 2025, such as the ongoing war in Ukraine. Risk management for power utilities, energy-related companies, and governments is becoming not only important but essential. This involves reliable forecasts. In risk management, probabilistic/interval forecasts are of particular interest, as point forecasts ignore information about uncertainty arising from model choice, data, or parameter calibration.

In the electricity price forecasting literature, several authors have focused on developing (probabilistic) forecasting models for power prices. See, e.g., [1], [2], and [3] for overviews. For general overviews of probabilistic/interval forecasting, see [4] and [5].

Despite various refined models, existing research often addresses data-related uncertainty, whereas uncertainty arising from model choice and calibration is often overlooked (see Section II-A for more details). Including this uncertainty, e.g., via ensemble or Bayesian approaches, considerably increases complexity and computational challenges, and it remains to

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be verified whether the additional effort pays off. So far, to the authors' best knowledge, there has been no deeper analysis in electricity price forecasting with regard to this question. Hence, we propose to compare probabilistic forecasts and prediction intervals (PIs) generated by distributional deep learning and statistical models in combination with various uncertainty quantification methods. In addition, we discuss which evaluation measure is adequate here. For illustration and testing purposes, we use historical energy price data from the German market. These data not only show high volatility but also multiple structural breaks in the past and are therefore adequate for thoroughly testing methods for uncertainty quantification and comparison with forecasts from other models. In this work, we demonstrate that accounting for both data and model uncertainty, particularly for deep learning models, yields significant benefits. As shown in Section IV, this approach leads to more reliable PIs as well as improved point forecasts. We also find that statistical models are highly competitive, even when compared to complex machine learning models. The main findings are summarized as follows:

- Uncertainty quantification benefits the performance of deep distributional neural networks in forecasting German electricity prices. Distributional accuracy is increased by up to 7%, and the mean absolute error of mean forecasts is decreased by up to 7%.
- For linear statistical models, different approaches to uncertainty quantification (parametric, ensemble or conformal) yield comparable results in terms of distributional accuracy, while conformal prediction captures uncertainty itself best.
- For the German market, the overall best performing model is the conformal-prediction XGBoost model.

The article is structured as follows: In Section II, we discuss state-of-the-art approaches for (probabilistic/interval) electricity price forecasting from both machine learning and statistics, commenting on their approaches to quantifying uncertainty and motivating additional approaches to uncertainty quantification. In Section III, the data for the case study is introduced. The forecasting study is presented in Section IV, where we discuss how models are calibrated and trained (including hyperparameter choices), and compare results using suitable metrics. Before we conclude the article in Section VI, we discuss our results in Section V.

II. A REVISION OF CONCEPTS FOR FORECASTING AND UNCERTAINTY QUANTIFICATION

There are various approaches for generating probabilistic and interval forecasts. First, there are parametric distribution-based approaches that rely on certain assumptions about the underlying probability distribution of the data. The PIs are constructed by estimating the parameters of the (conditional) distribution and using the corresponding quantiles, as in distributional deep neural networks (DDNNs) [6]. Then, some methods directly learn specific aspects of a forecast, e.g., direct quantile learning methods such as quantile regression [7], which do not require distributional assumptions. Finally, there are model-agnostic approaches such as bootstrapping [8], historical simulation [9], or conformal prediction [9], which do not rely on any specific underlying distribution or regression model. These approaches are applied to already trained models.

In Section II-A, we first review concepts of uncertainty quantification. Then we proceed to introduce state-of-the-art statistical and machine learning models for probabilistic electricity price forecasting and discuss them in the context of uncertainty.

A. A FEW NOTES ON UNCERTAINTY

Forecasting intervals are an attempt to quantify and estimate the uncertainty connected to the behavior of a stochastic process. However, uncertainty itself needs to be taken into account when designing or choosing a model for the generation of PIs.

Generally speaking, uncertainty can be categorized into two types: aleatoric uncertainty, which describes data-related uncertainty [10], and epistemic uncertainty, which relates to model-specific uncertainties [11], [12]. Aleatoric uncertainty arises from the variability inherent in the input data and cannot be reduced, not even by collecting more data. Sources of aleatoric uncertainty include measurement errors or the use of inherently uncertain input features, such as energy demand estimates or weather forecasts. Epistemic uncertainty, on the other hand, arises from uncertainties related to the choice of model features, such as the general architecture (e.g., using a linear model to capture nonlinear relationships), the parameters, and the distributional properties. Especially when dealing with nonstationary processes and potential structural breaks, we must address domain-shift uncertainty, which describes potential differences in the distributional properties of calibration and test data. Unlike aleatoric uncertainty, epistemic uncertainty can be reduced. Key strategies for minimizing it include selecting a suitable model architecture, employing an effective optimization and training process, and using a well-distributed and representative training dataset. Ultimately, the resulting overall uncertainty related to the prediction for a specific sample is called predictive uncertainty.

B. STATISTICAL APPROACHES

The standard approach to statistical interval forecasting is to assume an underlying distribution and obtain the interval by extracting the corresponding quantiles. The simplest way to model a distribution is the naive model. The model naively uses a previous days price as mean forecast. To estimate price distributions, historical simulation may be applied, relying on the widely used Gaussian distribution. This involves fitting distribution parameters to the forecast errors from a selected dataset. We refer to these models as Naive-HS$_{train}$ and Naive-HS$_{val}$, depending on whether the training or validation set was used.

With regard to the distribution's volatility parameters, there are more complex approaches, such as the generalized au-

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toregressive conditional heteroskedasticity (GARCH) model (see [13]), which models the volatility parametrically, using an autoregressive model for it. Other parametric approaches such as copulas, see e.g. [14], might be used as well but are beyond the scope of this paper. The uncertainty may also be captured by ensemble methods or conformal prediction.

We consider two sophisticated models for mean forecasting. These models are the LASSO estimated AR model (LEAR), cf. [15], and the extreme gradient boosting (XGBoost), cf. [16]. We augment these two mean-forecasting models with the following mechanisms to capture uncertainty. 1) applying quantile regression averaging to different mean predictions, i.e., XGBoost/LEAR-QRA. LEAR-QRA was proposed by [2] (see also [6], Sect. 4.3.2), but the methodology can be translated seamlessly to XGBoost. 2) fitting a GARCH model on the residuals of the XGBoost/LEAR model to obtain predictions for the conditional variance, which together with the conditional mean forecast parametrizes a conditional distribution. 3) Using conformal prediction on the mean forecasts of the respective models. We start by introducing the LEAR and XGBoost models for mean forecasting. Afterwards, the uncertainty modeling augmentations are described. The LEAR model for electricity price forecasting, originally introduced by [15], assumes an autoregressive model with external regressors for each product individually and then estimates the coefficients using a LASSO penalty (see [17] for the LASSO penalty). For each product, the model may be written as

$$y_t = \beta_0 + Y_t\beta + X_t\gamma + \varepsilon_t, \tag{1}$$

where $\{y_t\}_{t\in\mathbb{Z}}$ is the time series of interest, $Y_t = (y_{t-1}, \ldots, y_{t-p})$ contains past values of $\{y_t\}$, and $X_t = (x_{1,t-1}, \ldots, x_{1,t-p_1}, \ldots, x_{r,t-1}, \ldots, x_{r,t-p_r})$ contains the past values of the external regressors. The parameters are estimated by minimizing the LASSO-augmented sum of squares. If the penalty parameter is chosen correctly, the estimator will select only the relevant past lags of y_t and the external regressors. This enhances precision and decreases the variability of the mean predictions. Potential multicollinearity in the features is automatically taken care of.

As an alternative to the parametric LEAR model, we consider a tree based model in conjunction with boosting, i.e. exteme gradient boosting (XGBoost), cf. [16]. In the XGBoost framework, the idea of using boosting with trees as base learners, as in [18], is modified to be sparsity aware and computationally efficient. Problems such as multicollinearity are hence automatically taken care of by the model. Adjusted versions of XGBoost are used by e.g. [19] and [20] for electricity price forecasting. While it is structurally based on trees, it relies heavily in the statistical learning theory like gradient descent, regularization, etc, and hence we include XGBoost model in this section. Similar to the LEAR-model, XGBoost produces mean predictions. We augment the models by different approaches to capture uncertainty:

For the XGBoost/LEAR-QRA model, different mean predictions are required. We follow the procedure of [2] and train the models on different horizons to obtain different mean forecasts. In total m mean forecasts are obtained (in the application, we set $m = 4$). Then, a quantile regression with these mean forecasts as regressors is fitted for a range of percentile in order to generate PIs. To obtain probabilistic forecasts for an observation from the test data the models are re-fitted for the m horizons, and the respective mean predictions are obtained. Afterwards, the mean predictions are used in conjunction with the fitted quantile regression coefficients to obtain predictions for a range of percentile. The collection of the percentiles constitutes the probabilistic forecast.

The XGBoost/LEAR-GARCH model follows a different approach. Instead of relying on ensemble-based estimators for the probabilistic forecast, the conditional variance is predicted by a GARCH model. Assuming an underlying conditional distribution, the conditional mean (given by the LEAR prediction) and the conditional variance (given by the GARCH prediction) constitute a probabilistic forecast (in the application, we assume that the conditional distribution is the normal distribution). Last, in the XGBoost/LEAR-CP model, conformal prediction is applied to the mean-forecasts of the models (see Sect. II-D). While sharing a basic building block (the LEAR mean modeling), these approaches are fundamentally different. The GARCH approach quantifies uncertainty explicitly by modeling the conditional variance, whereas the LEAR-QRA approach captures uncertainty implicitly using an ensemble of predictors trained on different horizons. In particular, short-term training windows can account for recent increases in variance, as discussed in [2].

C. DEEP LEARNING-BASED CONCEPTS

As machine learning models are widely used today, we refrain from providing a detailed introduction to deep learning and machine learning models in general. Reader that are not familiar with such concepts are referred to [21].

Lago et al. [2] show that simple two-layer feed-forward neural networks are among the most effective deep learning models for point forecasting. Building on this, Marcjasz et al. [6] proposed the distributional deep neural network (DDNN), a probabilistic extension that estimates the parameters of the underlying price distribution rather than predicting electricity prices directly. The model is trained by minimizing the negative log-likelihood loss (NLL):

$$L_{NLL} = -\sum_{i=1}^{N} \log p(y_i \mid x_i; \theta), \tag{2}$$

where x_i and y_i are the inputs and observed targets, and $p(y_i \mid x_i; \theta)$ the probability assigned to y_i by the distribution predicted by the network with parameters θ. The output distribution captures data uncertainty in the predictions [10]. To additionally capture model uncertainty, various approaches can be used, including ensemble methods, Bayesian approaches such as Monte Carlo Dropout, and single deterministic methods such as Deep Evidential Regression [10].

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1) Ensembles

The idea of ensemble methods is to combine different models, i.e. their forecasts, to generate improved predictions, which come at a cost, as multiple models must be calibrated and maintained.

Ensembles also provide an intuitive way to measure model uncertainty by capturing the variation in predictions across the models. For NNs, it is essential to maximize the diversity of the network's behavior to obtain reliable uncertainty estimates (while maintaining good predictive performance). This can be done by training different architectures, training on different subsets of the original dataset, or varying the initial parameter set [22]. Lakshminarayanan et al. [23] found that using different random initializations of the NN weights and random shuffling suffices to create network diversity. To get the predictive distribution, one has to aggregate the ensemble models into a weighted mixture model:

$$p(y \mid x; \theta) \quad = \quad \sum_{j=1}^{M} w_j \, p(y \mid x; \theta_j), \tag{3}$$

using nonnegative weights $w_1, \ldots, w_M$ that add up to one. To calculate the quantile $Q(p;\theta) = F^{-1}(p;\theta)$ for a given probability $p \in [0,1]$, we solve the following equation:

$$F(x;\theta) - p = 0 \tag{4}$$

numerically. The initial guess for the solution we set to $x_0 = \sum_{i=1}^{M} w_j F_j^{-1}(p)$.

2) Monte Carlo Dropout

While ensemble models train several NNs to capture model uncertainty, Bayesian logic suggests an alternative approach by interpreting individual weights as probabilistic variables. The Monte Carlo (MC) dropout model is a widely used example [24]. It differs from the DDNN by including dropout layers after each hidden layer. These dropout layers remain active during both training and inference. As a result, the same input can produce different outputs in multiple forward passes through the network. During inference, we perform N forward passes and aggregate the results using a uniformly weighted mixture model. Quantiles are then obtained by solving Equation 4, equivalent to the ensemble approach.

3) Deep Evidential Regression

Deep evidential regression (DER) is a single deterministic method, introduced by Amini et al. [25], where one evidential deep neural network (EvDNN) predicts both the forecast and its uncertainty. In contrast to a Bayesian method, priors are placed directly over the parameters of the distribution instead of the network weights. If a Gaussian distribution is assumed, a Gaussian prior is placed on the mean and an inverse gamma prior on the variance:

$$\begin{aligned} y &\sim \mathcal{N}(\mu, \sigma^2), \\ \text{with } \mu &\sim \mathcal{N}(\gamma, \sigma^2 \nu^{-1}), \quad \sigma^2 \sim \Gamma^{-1}(\alpha, \beta), \end{aligned} \tag{5}$$

where $\gamma \in \mathbb{R}$, $\nu > 0$, $\alpha > 1$, and $\beta > 0$. This leads to a normal-inverse gamma distribution for the joint prior distribution, assuming that the parameters factorize as $p(\mu, \sigma^2) = p(\mu)(\sigma^2)$:

$$\begin{aligned} & p(\theta|m) \\ = \; & \frac{\beta^\alpha \sqrt{\nu}}{\Gamma(\alpha)\sqrt{2\pi\sigma^2}} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left(-\frac{2\beta + \nu(\gamma-\mu)^2}{2\sigma^2}\right) \\ = \; & NIG(\mu, \sigma^2; \gamma, \nu, \alpha, \beta) \end{aligned} \tag{6}$$

where we write $\theta = (\mu, \sigma^2)$ and $m = \gamma, \nu, \alpha, \beta$. Integrating over θ yields the predictive distribution for y:

$$\begin{aligned} p(y|m) &= \int p(y|\theta) p(\theta|m) \, d\theta \\ &= St\left(y; \gamma, \frac{\beta(1+\nu)}{\nu\alpha}; 2\alpha\right). \end{aligned} \tag{7}$$

This predictive distribution follows a Student's t-distribution with location γ, scale $\frac{\beta(1+\nu)}{\nu\alpha}$, and 2α degrees of freedom. The network is then trained by minimizing the NLL loss

$$\begin{aligned} L_{DER,NLL} &= \sum_{i=1}^{N} St\left(y_i; \hat{\gamma}_i, \frac{\hat{\beta}_i(1+\hat{\nu}_i)}{\hat{\nu}_i \hat{\alpha}_i}; 2\hat{\alpha}_i\right) \\ &= \sum_{i=1}^{N} \frac{1}{2}\log\left(\frac{\pi}{\hat{\nu}_i}\right) - \hat{\alpha}_i \log(\hat{\Omega}_i) \\ &\quad + \left(\hat{\alpha}_i + \frac{1}{2}\right)\log\left((y-\hat{\gamma}_i)^2 \hat{\nu}_i + \hat{\Omega}_i\right) \\ &\quad + \log\left(\frac{\Gamma(\hat{\alpha}_i)}{\Gamma\left(\hat{\alpha}_i + \frac{1}{2}\right)}\right), \end{aligned} \tag{8}$$

where $\hat{\Omega}_i = 2\hat{\beta}_i(1+\hat{\nu}_i)$, $\Gamma(\cdot)$ denotes the Gamma function, and $\hat{\gamma}_i$, $\hat{\nu}_i$, $\hat{\alpha}_i$, and $\hat{\beta}_i$ are the predicted parameters for each training sample (x_i, y_i). In addition to the NLL loss, an evidential regularizer is introduced to penalize predictions that are inaccurate while exhibiting low uncertainty. This term is defined as

$$L_{DER,R} = \sum_{i=1}^{N} |y_i - \hat{\gamma}_i| \cdot (2\hat{\nu}_i + \hat{\alpha}_i). \tag{9}$$

The overall loss function for the DER method is the sum of the NLL loss and the regularizer, weighted by a hyperparameter $\lambda_{DER} > 0$:

$$L_{DER} = L_{DER,NLL} + \lambda_{DER} L_{DER,R}. \tag{10}$$

D. MODEL-AGNOSTIC APPROACHES

Model-agnostic approaches do not require assumptions about a specific probability distribution or regression model. Instead, they can be applied on top of any forecasting model. Historical simulations are examples, as described in Section II-B. Another example is Conformal Prediction [9]. Given a trained model on historical data, PIs are constructed using nonconformity scores, computed from the residuals of a distinct calibration set for each hour of the day. For each day d

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and hour h, the absolute residual as the nonconformity score is computed as,

$$s_{d,h} = |\hat{y}_{d,h} - y_{d,h}|, \tag{11}$$

where $\hat{y}_{d,h}$ is the point prediction and $y_{d,h}$ is the observed electricity price at hour h of day d. For a desired quantile q and each hour h, we construct symmetric PIs as follows: For the lower quantile ($q < 0.5$):

$$\hat{q}^{(q)}_{d,h} = \hat{y}_{d,h} - Q_{1-2q}(S_{d,h}). \tag{12}$$

For the upper quantile ($q \geq 0.5$):

$$\hat{q}^{(q)}_{d,h} = \hat{y}_{d,h} + Q_{2q-1}(S_{d,h}). \tag{13}$$

where $Q_\alpha(S_{d,h})$ denotes the α-quantile of the hour-specific nonconformity scores $S_{d,h} = \{s_{d-n_{cal},h}, s_{d-n_{cal}+1,h}, \ldots, s_{d-1,h}\}$.

III. ELECTRICITY DATA

All data were retrieved from the SMARD platform (smard.de), operated by the German Federal Network Agency. For forecasting power prices, we consider not only the price itself but also load forecasts and renewable energy generation forecasts. Daily seasonality is captured using weekday dummies. The dataset covers the period from 1 October 2018 to 30 November 2024. However, entire days are regularly missing in the load forecast series for some particular days. The total dataset has $2,253$ daily entries, out of which only 65 contain missing data. It turns out that missing data does not correlate with any important events. All but 8 were due to Austria leaving the joint bidding zone with Germany and Luxemburg, which is a solitary event. The rows with missing data are removed from the dataset, resulting in a final dataset that begins on 12 October 2018. Daylight saving time adjustments are handled by interpolating the missing hour during the spring shift with the average of the neighboring hours and resolving the duplicated hour during the autumn shift by averaging the two values.

There may be strong correlation or even multicollinearity in the features, which could potentially cause instability in the predictors. However, since all models are subject to regularization, such problems are automatically taken care of.

The power prices we consider are hourly day-ahead prices with physical delivery, traded on the European Power Exchange (EPEX SPOT). The day-ahead auction is held daily, with price settlement taking place at noon for delivery on the following day [6].

An overview of the dataset is provided in Fig. 1, which shows several noteworthy characteristics. Negative prices appear regularly. However, the period between mid-2021 and December 2022 stands out as an exception, with almost no negative prices, comparatively high volatility, and considerably large price spikes. Price spikes can also be observed in other years and are due to the physical delivery of electricity, combined with the fact that, up to now, it cannot be stored in large quantities in Germany. As a consequence, in November 2024, when Germany experienced a period of low wind and limited solar generation, together with insufficient fossil power, this led to a price surge of more than 800 EUR per MWh. In general, price properties vary considerably over time, which is why we refrain from computing fundamental statistical measures such as skewness, variance, or kurtosis. Fig. 1 further supports the assumption of time-varying volatility.

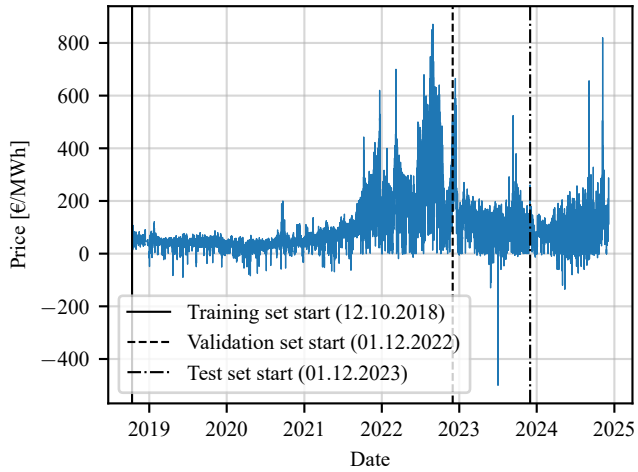

FIGURE 1. Day-ahead electricity prices from 12 October 2018 to 30 November 2024.

Both renewable energy generation and load forecasts are displayed in Fig. 2. Load exhibits clear annual seasonality, with lower demand in summer than in winter. While overall load volumes remain relatively stable across years, the structure within individual years is changing, though not to the extent of power prices. Forecasted renewable generation is increasing over time, consistent with infrastructure efforts in Germany. At this level, no distinct seasonality is visible, which may be explained by the negative correlation between forecasted solar and wind generation.

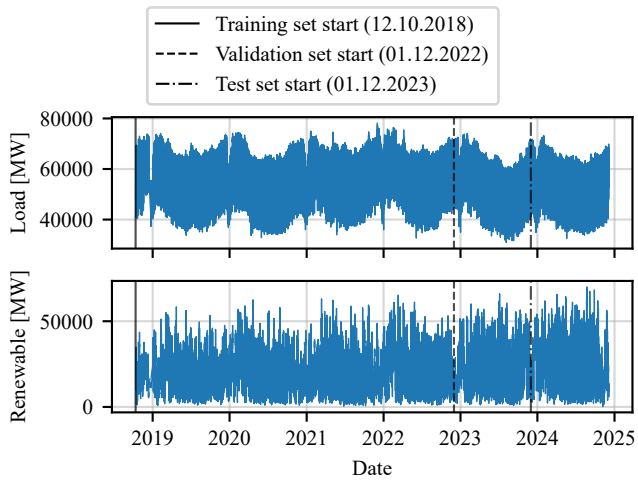

FIGURE 2. Load and renewable energy (wind + PV) forecasts from 12.10.2018 to 30.11.2024.

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Eventually, Fig. 3 presents scatter plots of load forecasts vs. power prices and renewable forecasts vs. power prices. Across the dataset, there is no clear dependence structure visible. However, this does not necessarily indicate the absence of interdependence but rather suggests a time-varying dependence structure.

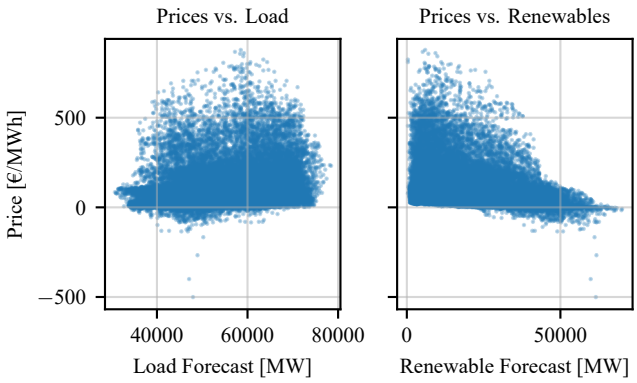

FIGURE 3. Load and renewable forecasts against day-ahead electricity prices from 12.10.2018 to 30.11.2024.

IV. CASE STUDY

To evaluate whether incorporating model uncertainty into deep learning models improves interval forecasts, we assess eight augmentations of the deep distributional neural network (DDNN) on the dataset described in Section III. These include ensemble models of 5 and 10 DDNNs (Ens5 and Ens10), MC Dropout models with 10 or 30 forward passes (MCD10 and MCD30), and an evidential deep neural network (EvDNN). Both ensembles and dropout models employ a Gaussian mixture aggregation scheme. In addition, all point forecasts (excluding Ens5 and MCD10) are combined with conformal prediction (suffix "–CP"). The DDNN, which accounts only for data uncertainty, serves as the baseline. For comparison, we also consider statistical benchmarks: LEAR-GARCH, LEAR-QRA, LEAR point forecasts with conformal prediction (LEAR-CP), and a naive approach combined with historical simulation based on the training and validation datasets (Naive-HS$_{train}$ and Naive-HS$_{val}$). In total, we evaluate 15 models. All of the models are implemented in the Python framework and they are publicly available through a Github repository[1].

A. INPUT FEATURES AND PREPROCESSING

The following 151 features are used as inputs to our models:

- Day-ahead prices from the past three days and from the same weekday of the previous week.
- Day-ahead forecasts of total load.
- Day-ahead forecasts of renewable energy generation.
- Weekday dummies, represented as binary vectors (e.g., [1, 0, 0, 0, 0, 0, 0] indicates Monday).

[1]Link to Python Code: Accessed on 1 Feb, 2026

Since the features have different scales, all 151 input features and 24 output values are standardized using area hyperbolic sine variance stabilization transformation [26]. Finally, the dataset is split into training, validation, and test sets. The training set covers 12 October 2018 to 30 November 2022, the validation set spans 1 December 2022 to 30 November 2023, and the test set includes all days between 1 December 2023 and 30 November 2024.

B. MODEL CALIBRATION AND TRAINING

All deep learning models use the "He" initialization for the hidden layers and the "Glorot" initialization for the output layer. During training, early stopping is applied with a patience of 100 epochs, and the maximum number of training epochs is set to 2000. The models are optimized using the ADAM optimizer with a batch size of 32. To reduce overfitting, all models employ L_2 regularization. Additionally, 10 independent runs are performed for each model, and the results are averaged. To ensure reproducibility, a fixed random seed is used for each training run. During these runs, the hyperparameters are fixed to the optimal ones, which are determined before. For the MCD10 and MCD30 models, only a single forward pass is performed during training, while validation involves 10 and 30 forward passes, respectively. To improve numerical stability during training, we follow the approach of [27] and predict the logarithm of the variance, $\log \sigma^2$, instead of predicting the variance σ^2 directly. For the EvDNN, we similarly predict $\log \nu$, $\log(\alpha - 1)$, and $\log \beta$ instead of ν, α, and β, and for the DDNN we predict $\log \sigma^2$ in place of σ^2.

For the LEAR model, we follow [3] and train for different length of datasets, specifically 56, 84, 1092, and 1461 days for 24 hour forecasting horizon. We average the results to produce the final point forecast. Additionally, we employ a rolling window scheme. The QRA and GARCH approaches use the mean forecasts obtained from the LEAR model based on the validation set. We fit 24 hourly GARCH models, each trained only on data from its corresponding hour. For the naive model, we use the previous day's price as the mean forecast for days during Tuesday and Friday, and prices from the same day of the previous week to predict prices for Monday, Saturday, and Sunday.

For conformal prediction, we use the validation samples as a calibration set, following [9]. For each test day d and hour h, we apply a rolling window of the most recent nonconformity scores specific to that hour, ensuring that only scores from the same hour across different days are used.

C. HYPERPARAMETER OPTIMIZATION

Before training, we performed hyperparameter optimization using the tree-structured Parzen estimator, a Bayesian optimization method implemented in the Optuna framework [28].

For the deep learning models, we optimized the learning rate, regularization parameters, dropout rate, and the evidence regularization parameter, with their corresponding search ranges listed in Table 4. We kept the network ar-

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chitecture fixed across all deep learning models to ensure comparability and to focus the benchmark on differences in uncertainty/interval prediction. Therefore, all deep learning models use the same architecture with two hidden layers of 1,024 neurons each. The hidden layers use ReLU activation functions, and the output layer uses a linear activation. Two-layer feed-forward networks are commonly used as strong baselines for day-ahead electricity price forecasting [29]. The choice of 1,024 neurons per layer follows the upper end of the width range considered in the extensive hyperparameter search of [30]. All hyperparameters, except the dropout rate, were sampled on a logarithmic scale. The optimization process consisted of 200 trials. The validation loss was used as the objective value for the tree-structured Parzen estimator algorithm. In each trial for the NN models, three runs were trained using the same hyperparameter values to reduce the effect of randomness, and the average validation loss across these three runs was then used as the objective value. Note that once the optimal hyperparameters are found, the actual training of ensembles of NNs is performed using these optimal hyperparameters.

For the LEAR and XGBoost models, we optimized the LASSO penalty parameter for LEAR as well as the XGBoost hyperparameters described in Table 4 using only the specification with the largest calibration window (1,461 days). The optimization process likewise consisted of 200 trials, with validation loss used as the objective. The resulting parameter settings were applied across all forecasting horizons.

D. ERROR METRICS

To measure point prediction accuracy, both the mean absolute error (MAE) and the root mean square error (RMSE) are used, with

$$\begin{aligned} MAE &= \frac{1}{24N_d}\sum_{d=1}^{N_d}\sum_{h=1}^{24}|p_{d,h}-\hat{p}_{d,h}|\,, \\ RMSE &= \sqrt{\frac{1}{24N_d}\sum_{d=1}^{N_d}\sum_{h=1}^{24}(p_{d,h}-\hat{p}_{d,h})^2}, \end{aligned}$$

where $p_{d,h}$ and $\hat{p}_{d,h}$ denote the observed and predicted prices at day d and hour h, respectively, and N_d is the number of prediction days.

For interval forecasts, we use the prediction interval coverage probability (PICP), the mean absolute average coverage error (MAACE), the continuous ranked probability score (CRPS; see also [31]), and the mean prediction interval width (MPIW). The PICP is defined as

$$PICP_{1-\alpha} = \frac{1}{N_{d,h}}\sum_{i=1}^{N_{d,h}} I_{d,h}^{(1-\alpha)},$$

$$I_{d,h}^{(1-\alpha)} = \begin{cases} 1 & \text{if } p_{d,h} \in [L_{d,h}^{(1-\alpha)},\ U_{d,h}^{(1-\alpha)}], \\ 0 & \text{otherwise,} \end{cases}$$

with $L_{d,h}^{(1-\alpha)}$ and $U_{d,h}^{(1-\alpha)}$ denoting the lower and upper bounds of the $(1-\alpha)$ PI [7]. The PICP measures the proportion of actual values that fall within the predicted interval. Moreover, to evaluate the overall calibration of a forecasting model across a range of confidence levels, we define *MAACE*:

$$MAACE = \frac{1}{N_\alpha}\sum_{\alpha\in\mathcal{A}}|ACE_{1-\alpha}|,$$

where $ACE_{1-\alpha} = PICP_{1-\alpha} - PINC_{1-\alpha}$ is the average calibration error, and *PINC* is the prediction interval nominal coverage, which is given as the deviation between the PICP and the nominal level, cf. [7]. The sharpness of the predicted quantiles is measured using the *CRPS*, which we calculate using the pinball loss function, [7],

$$\begin{aligned} CRPS_{d,h} &= \int_{-\infty}^{\infty}(\hat{F}(x)-\mathbb{1}_{\{p_{d,h}\le x\}})^2dx = \int_0^1 PS_{d,h}^q dq, \\ &\approx \frac{2}{99}\sum_{q=1}^{99} PS_{d,h}^q \end{aligned}$$

with

$$PS_{d,h}^q = \begin{cases} (1-q)(\hat{p}_{d,h}^q - p_{d,h}), & \text{if } p_{d,h} < \hat{p}_{d,h}^q, \\ q(p_{d,h}-\hat{p}_{d,h}^q), & \text{otherwise} \end{cases}$$

where $\hat{p}_{d,h}^q$ denotes the forecast of the q-th quantile of $p_{d,h}$. The scaling factor of 2 is commonly omitted in practice [9], a convention we follow here. The MPIW is defined as [32],

$$MPIW = \frac{1}{N_{d,h}}\sum_{i=1}^{N_{d,h}} U_{d,h}^{(1-\alpha)} - L_{d,h}^{(1-\alpha)}.$$

We compare forecasting performances of models pairwise using the Diebold-Mariano (DM) test for overall performance comparison and the Giacomini-White (GW) test for conditional comparisons, see [33] and [34] respectively, using the CRPS as the loss function. For the mathematical formulation of the DM test, please refer to Appendix B. Finally, we use the probability integral transformation (PIT) as a diagnostic tool to descriptively access the distributional fit of selected models. To this end, we evaluate the predicted distribution function for day d and hour h, $\hat{F}_{d,h}$, at the observation $p_{d,h}$. If the predicted distribution is well calibrated, the transformed variable, i.e. $\hat{F}_{d,h}(p_{d,h})$, ought to be standard uniformly distributed. In practice, the histogram of $\{\hat{F}_{d,h}(p_{d,h})\}_{d=\mathcal{I}}$ for some prediction interval $\mathcal{I}$, is inspected for deviations from uniformity. Since most of our probabilistic predictions are characterized by quantiles, we approximate the PIT-values by interpolating between percentiles.

E. ERROR ANALYSIS

We begin by evaluating the point prediction performance of the models. According to Table 1, the XGBoost model achieves the best overall point prediction performance, with an MAE of 12.020 and an RMSE of 20.534, outperforming all other models. The performance of LEAR models are significantly better than the deep learning models however they are outperformed by the XGBoost models by a small margin. As expected, the naive models yield significantly

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TABLE 1. Models are evaluated using point and probabilistic metrics. Results are averaged over 10 independent runs, with standard deviations shown in light gray. The best-performing results are highlighted in bold.

	MAE	RMSE	$\text{CRPS}_{1:99:1}$	$\text{MAACE}_{2:98:2}$
Naive-HS_{train}	24.878±0.000	36.685±0.000	9.823±0.000	12.842±0.000
Naive-HS_{val}	24.878±0.000	36.685±0.000	9.634±0.000	9.107±0.000
DDNN	17.209±0.767	25.684±0.913	6.392±0.286	5.286±2.735
Ens5	16.692±0.464	25.193±0.685	6.276±0.501	2.398±1.080
Ens10	16.577±0.211	24.942±0.262	6.087±0.069	2.370±1.043
MCD10	16.193±0.427	24.549±0.497	5.918±0.163	2.276±1.455
MCD30	16.116±0.412	24.426±0.447	5.897±0.149	1.962±1.379
EvDNN	15.940±0.697	24.566±0.674	7.510±0.311	4.937±0.730
DDNN-CP	17.209±0.767	25.684±0.913	6.330±0.271	1.948±0.361
Ens10-CP	16.577±0.211	24.942±0.262	6.114±0.078	2.272±0.126
MCD30-CP	16.116±0.412	24.426±0.447	5.942±0.141	1.736±0.568
EvDNN-CP	15.940±0.697	24.566±0.674	5.905±0.240	2.066±0.215
LEAR-GARCH	12.671±0.000	21.414±0.000	4.662±0.000	3.317±0.000
LEAR-QRA	12.671±0.000	21.414±0.000	4.670±0.000	2.865±0.000
LEAR-CP	12.671±0.000	21.414±0.000	4.666±0.000	1.305±0.000
XGBoost-GARCH	**12.020**±0.000	**20.534**±0.000	**4.492**±0.000	5.100±0.000
XGBoost-QRA	**12.020**±0.000	**20.534**±0.000	4.503±0.000	3.265±0.000
XGBoost-CP	**12.020**±0.000	**20.534**±0.000	4.493±0.000	**1.090**±0.000

higher errors compared to the NN-based and statistical approaches. Among the neural networks, the DDNN already achieves a considerable reduction in MAE and RMSE relative to the naive models. The ensemble models further improve performance, reducing MAE and RMSE compared to the DDNN. While increasing the number of models in the ensemble brings only marginal performance improvements, it noticeably reduces the standard deviation of the error metrics. MC Dropout models lead to even greater improvements, with reductions in MAE and RMSE of up to 6.4% and 4.9% relative to the DDNN. The number of forward passes in the MC Dropout models shows only a marginal positive effect on performance. Notably, the EvDNN achieves the lowest MAE among the neural models, with a 7.4% reduction compared to the DDNN.

When examining the interval forecasts, it becomes clear that the EvDNN model struggles to capture uncertainty accurately. It produces a relatively high MAACE of about 44% and the third-highest CRPS among all models. Similar to the point forecasts, the DDNN model significantly improves the probabilistic metrics compared to the naive baselines, and both ensemble and MC Dropout extensions further enhance performance. Specifically, they reduce the CRPS by about 4% and 7%, respectively, compared to the DDNN. For MAACE, these methods achieve notable improvements as well. The ensemble reduces MAACE by 3 points (a 49% improvement) and MC Dropout by 3.6 points (a 67% improvement), again relative to the DDNN. Interestingly, increasing the number of ensemble models has a strong impact on CRPS but little effect on MAACE, while the opposite trend is observed with the number of forward passes in the MC Dropout models. Another point worth noting is the relatively high standard deviation of MAACE produced by the DDNN, which is considerably larger than that of the other models. The XGBoost-GARCH model achieves the best overall forecasting performance in terms of CRPS, with a CRPS of 4.492, followed closely by the XGBoost-CP model at 4.493 and the XGBoost-QRA model at 4.503. When assessing the uncertainty measure MAACE, the top-performing model is the GBoost-CP with a value of 1.090, followed by the LEAR-CP at 1.305 and the MCD30-CP at 1.736.

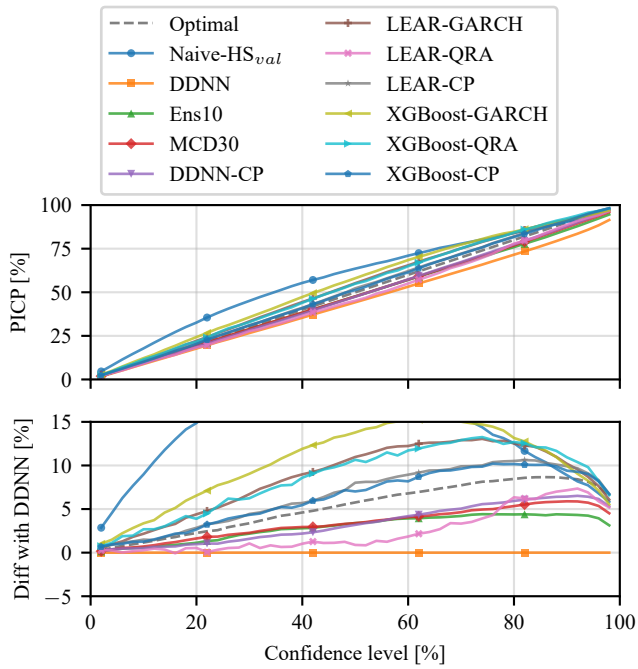

FIGURE 4. Models are evaluated using PICP across confidence intervals ranging from 2% to 98%. The top subplot shows the average results over 10 independent runs, with standard deviations indicated by dotted lines. The bottom subplot displays the difference in performance between the DDNN and the other models.

Next, we examine the PICP metric across different confidence levels to assess how well the predicted intervals are calibrated. The corresponding results are presented in Table

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TABLE 2. Models are evaluated using PICP at confidence intervals of 50%, 60%, 70%, 80%, 90% and 98%. Results are averaged over 10 independent runs, with standard deviations shown in light gray.The best-performing results are highlighted in bold.

[%]	$PICP_{50\%}$	$PICP_{60\%}$	$PICP_{70\%}$	$PICP_{80\%}$	$PICP_{90\%}$	$PICP_{98\%}$
Naive-HS_{train}	67.93 ±0.00	75.20 ±0.00	81.88 ±0.00	87.65 ±0.00	93.89 ±0.00	98.82 ±0.00
Naive-HS_{val}	63.55 ±0.00	71.00 ±0.00	77.87 ±0.00	84.10 ±0.00	**90.36** ±0.00	97.29 ±0.00
DDNN	44.24 ±3.51	53.21 ±3.83	62.28 ±3.91	71.54 ±4.02	81.47 ±3.57	91.47 ±2.37
Ens5	47.67 ±1.50	57.09 ±1.48	66.58 ±1.55	75.89 ±1.41	85.60 ±1.16	94.35 ±0.53
Ens10	47.63 ±1.37	57.13 ±1.45	66.48 ±1.58	75.93 ±1.57	85.83 ±1.33	94.60 ±0.81
MCD10	47.34 ±1.91	56.85 ±2.16	66.49 ±2.28	76.47 ±2.21	86.90 ±1.88	95.71 ±0.88
MCD30	47.63 ±1.65	57.27 ±1.96	66.92 ±2.16	76.89 ±2.02	87.36 ±1.86	95.98 ±0.85
EvDNN	4.84 ±0.53	6.13 ±0.76	7.63 ±0.97	9.82 ±1.23	13.42 ±1.72	22.17 ±2.81
DDNN-CP	47.43 ±0.48	57.39 ±0.51	67.32 ±0.48	77.48 ±0.47	87.90 ±0.36	96.78 ±0.13
Ens10-CP	46.98 ±0.18	57.15 ±0.19	67.17 ±0.21	77.31 ±0.25	87.66 ±0.14	96.66 ±0.08
MCD30-CP	47.53 ±0.71	57.31 ±0.98	67.39 ±0.91	77.59 ±0.84	88.21 ±0.57	96.99 ±0.15
EvDNN-CP	47.21 ±0.40	57.02 ±0.41	67.01 ±0.39	77.18 ±0.36	87.65 ±0.22	96.58 ±0.11
LEAR-GARCH	55.21 ±0.00	65.44 ±0.00	75.18 ±0.00	84.07 ±0.00	91.67 ±0.00	97.55 ±0.00
LEAR-QRA	45.13 ±0.00	55.05 ±0.00	65.61 ±0.00	77.89 ±0.00	88.63 ±0.00	96.65 ±0.00
LEAR-CP	52.27 ±0.00	62.15 ±0.00	72.18 ±0.00	82.06 ±0.00	91.38 ±0.00	**98.18** ±0.00
XGBoost-GARCH	58.03 ±0.00	68.48 ±0.00	77.30 ±0.00	84.70 ±0.00	91.44 ±0.00	97.01 ±0.00
XGBoost-QRA	54.90 ±0.00	64.92 ±0.00	75.18 ±0.00	84.23 ±0.00	92.41 ±0.00	98.06 ±0.00
XGBoost-CP	**51.26** ±0.00	**61.38** ±0.00	**71.89** ±0.00	**81.71** ±0.00	90.87 ±0.00	98.20 ±0.00

2 and Fig. 4. It should be noted that, of all the figures, there are 10 models under consideration, whereas the tables in this section include 15 models in total. Five models, namely Naive-HS_{train}, Ens5, MCD10, EvDNN, and EvDNN-CP, are not shown in Fig. 4 and Fig. 5, since we consider only one better-performing model from each category, and the EvDNN model struggles to capture uncertainty. The naive models consistently produce PICP values higher than the expected nominal coverage, indicating underconfident intervals. The same holds for LEAR-GARCH and LEAR-CP. In contrast, all deep learning models typically exhibit overconfidence in their PIs, a limitation shared by XGBoost-GARCH and XGBoost-QRA. However, these XGBoost variants demonstrate significant performance improvements specifically at confidence levels exceeding 80%. Ultimately, XGBoost-CP proves to be the most robust approach, outperforming all other models by achieving the best overall Prediction Interval Coverage Probability (PICP).

Both the ensemble and MC Dropout models show improved PICP values compared to the DDNN, especially as the confidence level increases. The greatest improvements are observed around the 90% confidence level. Up to the 60% confidence level, the performance of the ensemble and MC Dropout models is quite similar. Moreover, both the ensemble and MC Dropout models significantly reduce the standard deviation of results across independent runs by about half compared to the DDNN, indicating more stable performance. After the 60% confidence level, the DDNN-CP models surpass both the ensemble and MC Dropout models. At about the 80% confidence level, the LEAR-QRA begins to surpass the previous models, while the LEAR-CP achieves the best overall coverage, as reflected in the MAACE metric.

As shown in Table 3 and Figure 5, the MPIW increases almost linearly with the confidence level up to around the 70% confidence interval. As expected, the naive model produces significantly wider PIs on average, consistent with its high PICP values. In contrast, the remaining approaches, including the LEAR-based, XGBoost-based, and NN-based methods, generate considerably narrower intervals, roughly half the width of those from the naive model. The exception is the EvDNN model, which produces very sharp but less reliable PIs. Among the classical post-processing approaches, the QRA variants (for both LEAR and XGBoost) are designed to deliver sharp quantiles, and this is reflected in comparatively narrow PIs over low-to-mid confidence levels; in particular, LEAR-QRA consistently produces the narrowest PIs across all confidence levels among the traditional and NN-based models (excluding the EvDNN), with MPIW values

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TABLE 3. Models are evaluated using MPIW at confidence intervals of 50%, 60%, 70%, 80%, 90% and 98%. Results are averaged over 10 independent runs, with standard deviations shown in light gray. The best-performing results are highlighted in bold (EvDNN excluded).

[EUR]	$\text{MPIW}_{50\%}$	$\text{MPIW}_{60\%}$	$\text{MPIW}_{70\%}$	$\text{MPIW}_{80\%}$	$\text{MPIW}_{90\%}$	$\text{MPIW}_{98\%}$
Naive-HS$_{train}$	57.14 ±0.00	71.29 ±0.00	87.80 ±0.00	108.56 ±0.00	139.34 ±0.00	197.07 ±0.00
Naive-HS$_{val}$	49.94 ±0.00	62.32 ±0.00	76.74 ±0.00	94.89 ±0.00	121.79 ±0.00	172.25 ±0.00
DDNN	24.10 ±2.34	30.07 ±2.92	37.03 ±3.60	45.78 ±4.45	58.76 ±5.71	83.11 ±8.08
Ens5	26.20 ±3.95	32.31 ±3.48	39.86 ±3.73	52.84 ±9.92	75.12 ±23.83	103.55 ±24.82
Ens10	24.88 ±0.91	31.14 ±1.18	38.54 ±1.52	48.07 ±2.07	62.90 ±3.30	94.10 ±7.87
MCD10	24.41 ±1.02	30.55 ±1.28	37.77 ±1.58	47.00 ±1.98	61.01 ±2.59	88.59 ±3.83
MCD30	24.52 ±1.00	30.69 ±1.25	37.96 ±1.55	47.25 ±1.92	61.39 ±2.49	89.42 ±3.62
EvDNN	2.26 ±0.30	2.87 ±0.38	3.62 ±0.49	4.64 ±0.65	6.41 ±0.94	11.08 ±1.90
DDNN-CP	24.64 ±1.19	31.27 ±1.39	39.51 ±1.70	50.41 ±1.95	66.66 ±2.41	100.51 ±4.34
Ens10-CP	23.08 ±0.32	29.53 ±0.40	37.60 ±0.56	48.40 ±0.74	64.03 ±1.01	96.02 ±1.48
MCD30-CP	23.05 ±0.71	29.27 ±0.77	37.06 ±0.87	47.21 ±0.93	62.91 ±0.83	95.61 ±1.16
EvDNN-CP	22.22 ±1.36	28.38 ±1.59	35.89 ±1.96	45.98 ±2.35	61.61 ±2.83	93.93 ±2.79
LEAR-GARCH	23.42 ±0.00	29.22 ±0.00	35.98 ±0.00	44.49 ±0.00	57.10 ±0.00	**80.76** ±0.00
LEAR-QRA	**17.88** ±0.00	**23.15** ±0.00	29.63 ±0.00	39.18 ±0.00	54.13 ±0.00	98.68 ±0.00
LEAR-CP	20.46 ±0.00	25.61 ±0.00	31.74 ±0.00	39.72 ±0.00	52.84 ±0.00	83.60 ±0.00
XGBoost-GARCH	22.78 ±0.00	28.43 ±0.00	35.01 ±0.00	43.29 ±0.00	55.56 ±0.00	78.58 ±0.00
XGBoost-QRA	19.77 ±0.00	25.06 ±0.00	31.86 ±0.00	41.97 ±0.00	58.67 ±0.00	107.46 ±0.00
XGBoost-CP	18.39 ±0.00	23.21 ±0.00	**29.13** ±0.00	**37.34** ±0.00	**51.91** ±0.00	90.30 ±0.00

of 17.88 at 50% confidence and about 39.18 at 80%. At higher confidence levels (notably 90% and 98%), however, the calibration-oriented variants become more competitive: the CP- and GARCH-based constructions (for both LEAR and XGBoost) tend to provide wider but more conservative intervals, and in this regime LEAR-CP and LEAR-GARCH perform best. These values are substantially lower than those of DDNN, which produces intervals of 24.10 and 45.78 at the 50% and 80% levels, respectively. The ensemble with 5 models shows a high standard deviation across runs, suggesting that its performance is sensitive to the specific composition of the ensemble. Up to the 50% confidence level, the MPIW values of the ensemble and MC Dropout models (excluding Ens5), as well as the DDNN, are quite similar; beyond this point, the ensemble and MC Dropout models begin to diverge from the DDNN as epistemic uncertainty becomes more influential in the tails. At the 90% confidence level, where the highest gains in PICP are observed, the Ens10 and MC30 models improve PICP by 4.36 and 5.89 percentage points, respectively, in exchange for increases in MPIW by 4.14 and 2.63, reflecting the trade-off between interval width and coverage. Between the two, MC Dropout performs better overall. Additionally, we calculate the probability integral transform (PIT) of the actual observations w.r.t. the predictive distributions of the respective models. If a models probabilistic forecast is well calibrated, the PIT-values follow a uniform distribution. To assess this, we plot the histograms of the PIT-values (Fig. 12 in the Appendix D) and inspect deviations from uniformity. It can be observed that the XGBoost model with conformal prediction uncertainty quantification is well calibrated. The DDNN model suffers from probability mass near the boundaries, i.e. 0 and 1. This can be explained by the model not being able to properly predict extreme events. The Monte Carlo dropout approach to the DDNN model seems to mitigate this problem at least for the upper boundary 1. Correcting the inflation near the boundary 0 is a potential future research direction.

F. STATISTICAL TESTS

In this section we discuss the results of applying two standard forecast comparison procedures: (a) the Diebold–Mariano test (DM test), see [33], and (b) the Giacomini–White test (GW test), see [34], for conditional predictive ability. In this and in the following section, we leave out the following five models from our analysis Naive-HS$_{train}$, Ens5, MCD10, EvDNN, and EvDNN-CP, since we only want to consider well-performing models from each category.We choose a significance level of $\alpha = 0.05$. We decide against multiple

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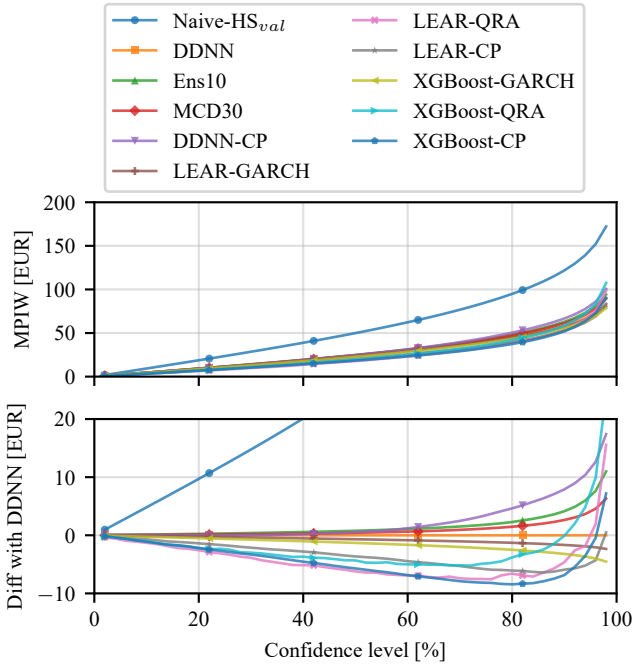

FIGURE 5. Models are evaluated using MPIW across confidence intervals ranging from 2% to 98%. The top subplot shows the average results over 10 independent runs, with standard deviations indicated by dotted lines. The bottom subplot displays the difference in performance between the DDNN and the other models.

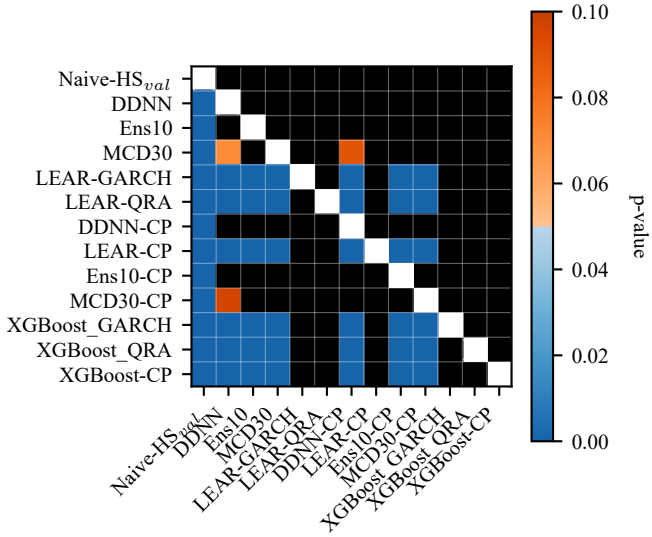

FIGURE 6. Heatmap of pairwise Diebold–Mariano test *p*-values using CRPS loss. Values below 0.05 mark significant differences in predictive accuracy. Results are averaged over 10 independent runs, yielding 130 comparisons per model pair across runs.

testing correction (e.g. via Bonferroni correction). and we do not apply multiple-testing correction (e.g., Bonferroni), so the results should be interpreted as exploratory pairwise evidence rather than familywise-controlled claims.

1) Diebold-Mariano Test

The p-values in the DM test show whether the two models under consideration are significantly different in terms of prediction accuracy. However, they do not provide any information about models performance superiority. To extract this information, one has to analyze the test statistic. The mathematical form of the DM test and the table of the test statistic are given in Appendix B in Table 8. Table 8 reports the corresponding DM statistics $d_{i,j}$ (based on CRPS loss), where $d_{i,j} < 0$ means M_i attains lower average loss than M_j (row is better), $d_{i,j} > 0$ means the opposite (column is better), $|d_{i,j}|$ quantifies the strength of the advantage, and values numerically near 0 indicate practical equivalence. In Fig. 6, we show the pairwise p-values (row model M_i vs. column model M_j), with significance level 0.05. Entries whose numeric values on the color bar are < 0.05 indicate that the observed difference between the two models is statistically significant, whereas larger p-values > 0.05 indicate no detectable difference. The diagonal elements are left blank since models are not tested against themselves.

Each model is compared with every other model across 10 independent runs. The results show that the three XGBoost–type models form a statistically tied top tier: XGBoost–GARCH, XGBoost–QRA, and XGBoost–CP performances are mutually indistinguishable, with near-zero differentials and non-rejections in all pairwise directions, while LEAR variants being competitive to XGBoost variants. The pairwise DM statistics given in Table 8 in Appendix D support this claim. For further clarity on the p-values, we also provide their numerical values in Table 9 in the Appendix D. However, each of these XGBoost models dominates every other models, except the LEAR-CP and LEAR-QRA, by a wide and statistically significant margin. All deep-learning models (DDNN, Ens10, MCD30) and their conformal/MC Dropout variants (DDNN-CP, Ens10-CP, MCD30-CP) are statistically indistinguishable from one another at $\alpha = 0.05$; however, directional tendencies do exist as MCD30 seems to be performing better, generally, although this difference is not significant. Conformal and MC Dropout models do not yield measurable gains (in terms of CRPS) over their corresponding base models. Finally, every method strictly improves upon the naive baseline.

In summary, (i) there is no single best model within the XGBoost family, but together they form a statistically superior performance frontier; (ii) deep-learning methods are statistically tied with one another and uniformly inferior to the XGBoost and LEAR models; and (iii) naive forecasting is decisively outperformed by all other approaches.

2) Giacomi - White Test

The Giacomini–White (GW) test evaluates conditional predictive ability rather than only unconditional mean performance. The key design choice in the GW framework is the information set on which predictive ability is evaluated. This is encoded through the instrument vector [c.f. Appendix C],

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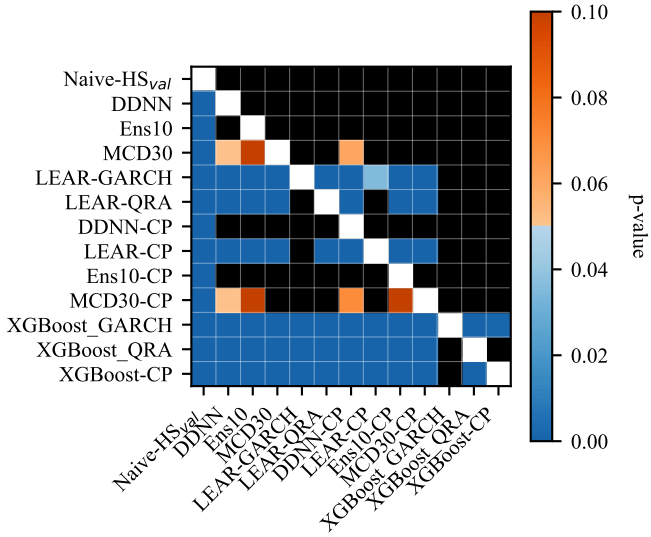

FIGURE 7. Heatmap of pairwise Giocami-White test *p*-values using CRPS loss. Values below 0.05 mark significant differences in predictive accuracy. Results are averaged over 10 independent runs, yielding 130 comparisons per model pair across runs.

which is required to be measurable with respect to the forecaster's information available at a given time. The mathematical description and the tables for the P-values and the test statistic are given in the Appendix C and D respectfully. Taking the motivation from [26], we implement the GW test and found that empirically, the GW statistics are extremely large for most pairs, implying essentially zero p-values under the χ^2_2 reference. For example, $GW_{\text{DDNN, Naive}} = 207.52$ corresponds to a p-value on the order of 10^{-46}, and $GW_{\text{XGBoost-GARCH, Naive}} = 710.53$ corresponds to a p-value on the order of 10^{-155}. This indicates overwhelming evidence that the models differ in conditional predictive ability relative to the naive baseline. Likewise, comparisons of the neural baselines against LEAR/XGBoost variants are associated with very large GW statistics (typically > 150 and often > 200), again implying p-values effectively zero and confirming that these families are not conditionally equivalent.

The striking difference compared to the DM test appears within the top-performing families. Although the DM statistics between LEAR variants are near zero (suggesting practical equivalence in mean CRPS), the corresponding GW statistics are not small. For example, $GW_{\text{LEAR-GARCH, LEAR-QRA}} = 151.41$ (p-value $\approx 1.3 \times 10^{-33}$) indicates strong rejection of conditional equality even when the unconditional mean gap is negligible by DM. The same phenomenon is visible for several other "close" model pairs: $GW_{\text{LEAR-GARCH, LEAR-CP}} = 6.65$ (p-value ≈ 0.036) and $GW_{\text{XGBoost-CP, XGBoost-GARCH}} = 12.44$ (p-value ≈ 0.002). These results mean that, under $z_t = (1, d_t)$, the loss differential does not behave like a conditionally mean-zero martingale difference; rather, it exhibits detectable temporal structure captured by d_t (persistence/reversal in relative performance), even when the average difference is small.

Hence, the GW test does not contradict the DM test; it answers a stronger question: not only whether the average CRPS values are equal, but also whether relative performance is conditionally stable when conditioned on the most recent loss differential. In our results, several model pairs that appear practically tied in mean CRPS under DM nonetheless show significant differences under GW, which is precisely the type of conditional structure that the additional instrument d_t is designed to detect.

G. TRADING STRATEGIES

A different angle to evaluate the goodness of fit of a specific forecasting method is to integrate it into a trading strategy and observe the financial success. We follow the suggestion of [35] and aim to optimally control a battery storage system based on interval forecasts of the German day-ahead power price. Let the battery have a capacity of $B = 2\,MWh$ and a charging/discharging efficiency of $\xi = 90\%$. Each trading day d, with $d = 1, \ldots, D$ and D being the number of trading days, we pick an hour h_{buy} for buying $1\,MWh$ and another hour h_{sell} for selling it. We start with an initial charge of $B_d = 1\,MWh$ on day $d = 1$. Transaction costs are ignored.

We consider a quantile-based strategy that uses PIs generated by each of the methods tested in Section IV-E, as well as the perfect foresight strategy, where we know the true prices, as the profit's upper boundary. The quantile-based strategy works by setting limits, which are given by the PIs: the selling price is equal to the lower boundary of the highest price's PI $L^{(1-\alpha)}_{d,h_{sell}}$; correspondingly, we obtain the purchasing price via the upper boundary of the lowest forecasted price's PI $U^{(1-\alpha)}_{d,h_{buy}}$ given a $1-\alpha$ confidence level. We perform both trades if and only if

$$\xi\, L^{(1-\alpha)}_{d,h_{sell}} - \frac{1}{\xi}\, U^{(1-\alpha)}_{d,h_{buy}} > 0, \quad h_{sell} \neq h_{buy}, \tag{14}$$

with h_{sell}/h_{buy} being the hour of the highest/lowest predicted price. Since we use limits, trading is not mandatory, and we might end a day with $B_d = 2\,MWh$ or $B_d = 0\,MWh$. In that situation, to avoid staying at the battery's margins, we place two limited and one unlimited bid the next day by solving the following optimization problem:

(a) If $B_d = 2MWh$: Solve

$$\max_{h_{sell,1}, h_{sell,2}, h_{buy}} \xi\, \hat{p}_{d,h_{sell,1}} + \xi\, \hat{p}_{d,h_{sell,2}} - \frac{1}{\xi}\, \hat{p}_{d,h_{buy}}, \tag{15}$$

$$h_{sell,1} \neq h_{sell,2} \neq h_{buy}, \quad h_{sell,1} < h_{buy}. \tag{16}$$

The trades are executed if:

$$\xi\, \hat{p}_{d,h_{sell,1}} + \xi\, L^{(1-\alpha)}_{d,h_{sell,2}} - \frac{1}{\xi}\, U^{(1-\alpha)}_{d,h_{buy}} > \xi\, \hat{p}_{d,h_{sell,3}}, \tag{17}$$

where $h_{sell,3}$ is the hour with the highest predicted price. If this condition is not met, an unlimited sell order is placed at $h_{sell,3}$.

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(b) If $B_d = 0MWh$: Solve

$$\max_{h_{sell},h_{buy,1},h_{buy,2}} -\frac{1}{\xi}\hat{p}_{d,h_{buy,1}} + \xi\,\hat{p}_{d,h_{sell}} - \frac{1}{\xi}\hat{p}_{d,h_{buy,2}}, \quad (18)$$

$$h_{sell} \neq h_{buy,1} \neq h_{buy,2}, \quad h_{buy,1} < h_{sell}. \quad (19)$$

The trades are executed if:

$$-\frac{1}{\xi}\hat{p}_{d,h_{buy,1}} + \xi\,L^{(1-\alpha)}_{d,h_{sell}} - \frac{1}{\xi}U^{(1-\alpha)}_{d,h_{buy,2}} > -\frac{1}{\xi}\hat{p}_{d,h_{buy,3}}, \quad (20)$$

where $h_{buy,3}$ is the hour with the lowest predicted price. If this condition is not met, an unlimited buy order is placed at $h_{buy,3}$.

To make sure we end at $B_d = 1$, we trade unilaterally if $B_d = 0$ or $B_d = 2$ on the last day of the trading period. Eventually, we evaluate performance based on the total profit over the test period and the per-transaction profit, which is calculated by dividing the total profit by the number of executed trades.

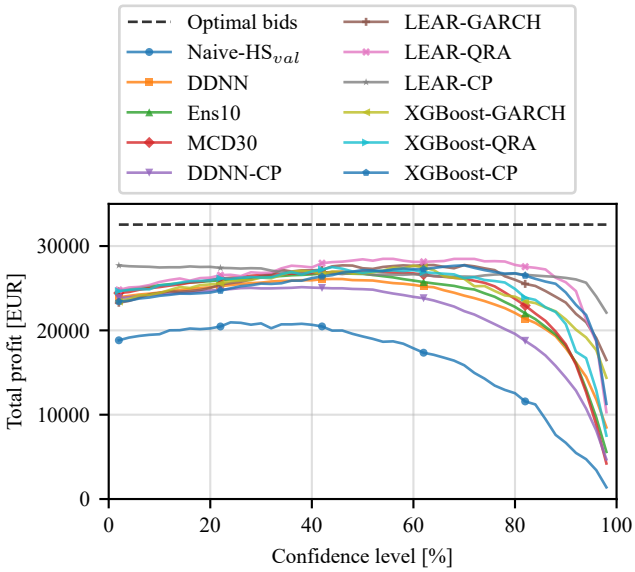

FIGURE 8. Total profit of models is shown for confidence intervals ranging from 2% to 98% using the quantile-based strategy. For comparison, the optimal bids strategy is also included. Results are averaged over 10 independent runs.

In Fig. 8, the total profits for all tested strategies are shown depending on the chosen confidence level. We see that each model's total payoff peaks at a certain confidence level, after which it declines, and this peak confidence level differs by model. The NN-based model's profit, for example, culminates around 50%, while the naive model peaks around 30%. For the LEAR-based models, the peak shifts to higher confidence levels (around 85% for LEAR-QRA and around 90% for LEAR-GARCH), indicating that these models can remain profitable even when the strategy becomes more conservative. For the XGBoost family, the peak typically occurs at medium confidence levels (roughly in the 60–70% range), with XGBoost-CP generally performing best among the XGBoost variants across a broad range of confidence levels.

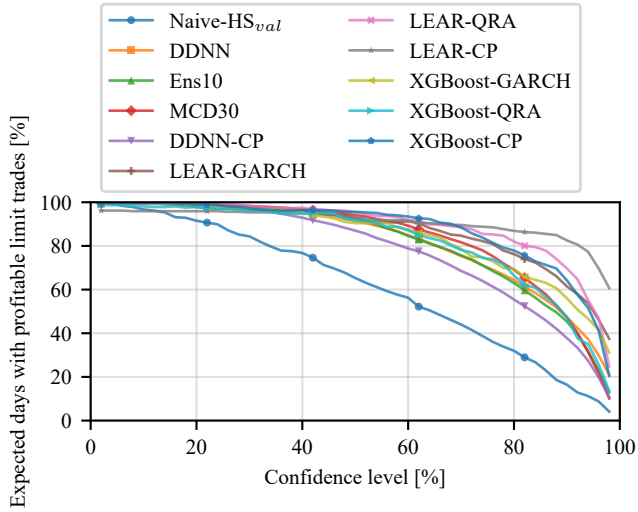

FIGURE 9. The expected number of days with profitable limit trades is shown for each model across confidence intervals ranging from 2% to 98% using the quantile-based strategy. Results are averaged over 10 independent runs.

The decline in total profit at higher confidence levels is caused by the widening PIs, which lead to fewer trading days identified as profitable and therefore fewer executed trades, as shown in Fig. 9. This mechanism applies to the XGBoost variants as well: at high confidence levels, the intervals become wider, the trading conditions are satisfied less frequently, and total profit drops even if the remaining trades can be individually attractive.

Both the ensemble and MC Dropout models outperform the DDNN model in terms of total profit at the 90% confidence level, despite having wider PIs as shown in Fig. 5. At the 70% confidence level, where the difference in total profit is most significant, the ensemble model achieves a profit that is 3.7% higher, while the MC Dropout model achieves an 8.4% higher profit compared to the DDNN model. In the same confidence region, the XGBoost models are competitive with the NN-based approaches, and the conformalized variant (XGBoost-CP) is typically the strongest within the XGBoost group, reflecting the benefit of improved calibration for the trading rule.

Notably, the LEAR-QRA model outperforms all other models between confidence levels of approximately 35% and 85%, until the PIs become too wide. Interestingly, below 35% and again between 85% and 95%, the LEAR-CP model achieves the highest total profits, showing that calibration can be especially valuable in the very aggressive (low confidence) and very conservative (high confidence) regimes.

Fig. 10 shows the profit per transaction. The per-transaction profit in quantile-based strategies tends to increase with higher confidence levels. In contrast, the statistical models show an initial decline up to around the 50% confidence interval, after which their profits begin to rise. Interestingly, the LEAR-CP model remains relatively stable up to around 90%, before it also starts to increase.

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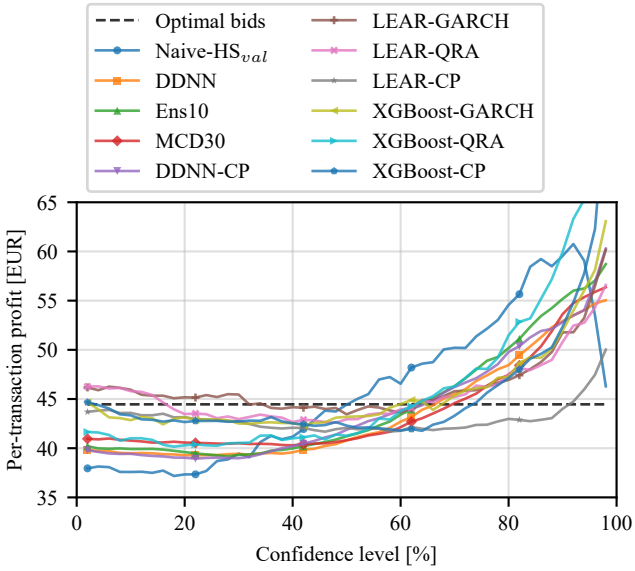

FIGURE 10. Per-transaction profit of models is shown for confidence intervals ranging from 2% to 98% using the quantile-based strategy. For comparison, the optimal bids strategy is also included. Results are averaged over 10 independent runs.

After reaching the 50% confidence level, the naive model begins to outperform the others in terms of per-transaction profit. This suggests that although it executes fewer trades, the trades it does make are generally more profitable; however, its profit curve starts to decline at the 90% confidence level. The XGBoost variants follow the same general logic of "fewer but stronger" trades at high confidence levels: as confidence increases, the strategy becomes more selective, which raises profit per executed trade even though total profit can still fall due to the reduced number of transactions.

Notably, the NN models and their CP counterparts exhibit very similar per-transaction profits. A similar pattern is visible for XGBoost and its conformalized version: differences in per-transaction profit are usually modest at medium confidence levels, while at very high confidence levels the per-transaction profit can rise sharply because only the most favorable opportunities remain tradable. Finally, the statistical models demonstrate higher per-transaction profits than all other models during the first half of the confidence level; afterward, LEAR-GARCH and LEAR-QRA tend toward the performance of the other models, while LEAR-CP remains significantly lower than the rest.

The tables for Fig. 8, Fig. 9, and Fig. 10, along with their corresponding plots against PICP, can be found in Appendix D.

V. DISCUSSION AND CRITICAL REFLECTION

In general, it shows that there is no superior method that performs best across all error measures. However, there is a certain tendency towards XGBoost and LEAR-based models, which show better performance in the presence of state-of-the-art deep learning benchmarks, despite being much simpler and requiring considerably less computational power. This indicates, at least given the setup of this case study, that the additional complexity of deep learning models is not paying off.

Our case study shows that the XGBoost-CP model has, on average, the most reliable PIs across all models, with a MAACE of 1.090, whereas the best deep learning model, MC Dropout, achieves a MAACE of 1.735. In contrast, the XGBoost-QRA and XGBoost-GARCH variants are noticeably less reliable in terms of MAACE, illustrating that strong point prediction alone does not guarantee well-calibrated intervals unless the uncertainty layer is appropriate.

At a confidence level of around 85% and higher, the LEAR-QRA starts to outperform the MD model, achieving a better PICP. The evidential neural network failed to capture the uncertainty. One possible reason could be the zero-gradient problem for evidential regression models; see [36] for more information. The LEAR-GARCH also had difficulties in providing reliable PIs, as did the naive approach; and a similar limitation is visible for the XGBoost-GARCH construction, whose uncertainty is entirely driven by residual dynamics and the chosen parametric distributional assumption. The problem is that these approaches rely heavily on how representative the historical residual period is for the current regime and on whether the assumed conditional distribution is appropriate. The superior performance of the XGBoost and LEAR-based models may be explained by the fact that they are refitted for every prediction. This is only possible because of the relatively low computational cost.

A possible reason could be the use of the rolling window scheme in contrast to the NN models, where the training data is fixed. In particular, both LEAR and XGBoost can be retrained frequently with updated windows, which allows them to adapt to changing levels of volatility and changing feature relevance; in contrast, the NN models in our setup are trained once on a fixed dataset and then evaluated, making them more exposed to nonstationarity and domain shifts unless additional adaptation mechanisms are introduced.

Our study indicates that quantifying both data and model uncertainty, particularly for deep learning models, pays off. In particular, ensemble-based methods and MC Dropout improve robustness and stability relative to a single DDNN, and often also improve point accuracy. Interestingly, evidential deep neural networks can achieve strong point accuracy but still generate less reliable PIs compared to their benchmarks.

We find that improvements in reliability come at the cost of wider PIs, i.e., reduced sharpness. From the perspective of the XGBoost family, this trade-off appears in a different form: XGBoost-QRA prioritizes sharpness by learning quantiles directly, XGBoost-GARCH prioritizes a volatility-based parametric uncertainty model, and XGBoost-CP prioritizes calibration through empirical error distributions; the empirical results indicate that the calibration-oriented approach is the most robust here.

Interestingly, incorporating model uncertainty into DDNN models also helps improve their point forecasts. This is due

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to the procedures to quantify and estimate model uncertainty, and not due to modeling uncertainty itself.

In terms of economic benefits, good performance in interval forecasting is no guarantee of actual profit in daily trading. Rather, the confidence level acts as a decision knob: as predicted limits become more conservative, the strategy becomes more selective, which typically reduces the number of trades but can increase per-transaction profit.

Eventually, there are a few limitations of the tested models that need to be mentioned. For example, we see that it pays off to use ensemble models for generating PIs. However, the choice of the required hyperparameters is rather tricky. Ensemble forecasting requires specifying the number of ensemble members, which is difficult to determine. Additionally, the composition of these members is significant. Weak members can harm overall performance, and equal-weight aggregation might be suboptimal.

In the statistical models, we implemented a rolling window approach for calibrating the parameters. Although this is beneficial in many situations, it causes problems during structural breaks. Moreover, the window size needs to be chosen beforehand, and this choice is always somewhat arbitrary. Statistical models are lighter and interpretable but can be brittle under regime shifts and heavy tails if distributional assumptions are too restrictive. ML models are adaptive but need careful regularization and calibration to avoid over- or underconfidence. Rather than choosing one, the evidence suggests a hybrid path: use LEAR-type models for baseline calibration and coverage control, use XGBoost-type models to capture nonlinear feature effects efficiently, and combine them with explicit calibration layers (such as CP) when diagnostic plots indicate systematic miscalibration.

In this study, Gaussian innovations were used for comparability. Moreover, a natural extension is to test heavy-tailed and skewed families (Student-t, skew-t, generalized error, generalized hyperbolic or JSU) within LEAR-GARCH. The same extension is relevant for the residual-based XGBoost-GARCH construction: if residuals are heavy-tailed or skewed, a Gaussian assumption can distort tail quantiles and thereby coverage at high nominal levels. This could retain the calibration advantages while improving tail coverage, again providing a way to strengthen statistical/parametric uncertainty layers without diminishing the role of ML.

Several mechanisms identified in this study are not country-specific but are shared by markets with high shares of variable renewable energy. However, generalizing our findings is hard. Whether our results can be transferred depends on various factors. Regulation, which for example might inhibit or ease market access, varies across countries. Systems with an equal or even larger share of renewable power such as Norway) may actually exhibit weaker price volatility as their hydro generation capacities function as buffer storages. Conversely, systems with high shares of non-dispatchable renewables but limited storage are likely to experience dynamics closer to those observed in Germany. The French market, still dominted by nuclear power, may show similar price patterns as, lately, price effects are often due to German power surplus supply or additional power demand.

VI. CONCLUSION

In this article, we test a wide range of methods on their capability to produce reliable interval forecasts for hourly power prices. In this context, we analyze whether forecasting accuracy improves if model and data uncertainty are explicitly considered using an adequate model, such as a Bayesian approach. An extensive model comparison based on various goodness-of-fit measures is performed, whereby both statistical and machine learning models are tested. It shows that there is no model that performs best across all measures. Moreover, having also considered economic implications in the context of a trading scheme, we see that increasing the precision of interval forecasts does not necessarily increase profits. With regard to prediction intervals, it pays off to account for uncertainty, and a combined focus on both aleatoric and epistemic uncertainty is beneficial for neural approaches. Ensemble models help to reduce both MAE and RMSE compared to a single DDNN and also stabilize results across runs. However, they are outperformed by classical baselines in this case study, where refitting and calibration play a central role. In particular, statistical LEAR models remain highly competitive, and combining LEAR with a GARCH volatility component proves to be fairly reliable for interval construction. Beyond LEAR, the tree-based XGBoost family provides a strong "middle ground" between linear statistical models and deep neural networks: it captures nonlinear feature interactions with modest training cost, and its performance depends crucially on the uncertainty layer applied on top of the mean model. In terms of distributional accuracy, the residual-based XGBoost–GARCH variant achieves the best overall CRPS in our study, and XGBoost–QRA yields similarly competitive CRPS values. In terms of reliability, however, the conformalized variant XGBoost–CP is the most robust approach, achieving the lowest MAACE and strong coverage behavior across nominal levels. This highlights that, even with a strong nonlinear mean predictor, well-calibrated intervals are not automatic; rather, calibration-oriented procedures such as conformal prediction can be decisive for producing reliable uncertainty estimates. From a statistical standpoint, the strength of the LEAR family is calibration. Their interval systems are directly tied to quantile/expectile targets, which often translates into well-behaved Probability Integral Transform (PIT) diagnostics and reliable Coverage vs. Nominal (CvN) curves. A constructive next step, without undermining ML, would be to use LEAR-based forecast distributions as calibration anchors for neural outputs (e.g., post-hoc quantile mapping or conformalized residuals). Eventually, we are aware of a few limitations of our study and paths for future research. First, only the German power market is tested, and it remains to be verified how the models perform on other markets. With regard to machine learning models, we could test hybrid uncertainty quantification methods, i.e., we could try to combine complementary methods in order

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to capture both epistemic and aleatoric uncertainty. A natural choice would be to create ensembles of MC Dropout models. Other hybrid approaches can be a combination of complementary methods to improve uncertainty estimation. For instance, we can combine distributional models with conformal calibration, which can fix miscalibration while preserving distribution shape, and deep ensembles combined with conformal prediction to capture model uncertainty while ensuring valid coverage. Post-hoc calibration is not yet fully explored [37]. Other methods, such as adaptive ensemble sizing and performance-based ensemble member selection, might also help to improve PIs. Furthermore, more complex distribution families, such as the generalized hyperbolic distribution for modeling randomness, could be explored, or alternatively, distribution-free approaches. Eventually, considering the comparably poor financial performance of the tested complex models, one needs to explore new possibilities that arise from the increasing availability of storage capacity on the German market. Here, uncertainty quantification could be more beneficial than in the currently tested scenario.

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APPENDIX.

A. LIST OF ABBREVIATIONS

ACE	average calibration error
CP	conformal prediction
CRPS	continuous ranked probability score
DER	deep evidential regression
DDNN	distributional deep neural network
DNN	deep neural network
DM	Diebold-Mariano
EvDNN	evidential deep neural network
GARCH	generalized autoregressive conditional heteroskedasticity
GW	Giocami-White
LASSO	least absolute shrinkage and selection operator
LEAR model	Lasso-estimated autoregressive model
LSTM network	Long short-term memory network
MAACE	mean absolute average coverage error
MAE	Mean absolute error
MC	Monte Carlo
MCD	Monte Carlo dropout
MPIW	mean prediction interval width
NLL	negative log likelihood loss
NN	neural network
PI	prediction interval
PICP	prediction interval coverage probability
PINC	prediction interval nominal coverage
PIT	probability integral transform
QRA	quantile regression averaging
RMSE	Root mean square error

B. THE DIEBOLD-MARIANO TEST

The Diebold-Mariano (DM) test is designed as follows: Let e_{1t} and e_{2t} denote the forecast errors of Model 1 and Model 2, respectively, at time t. To quantify the loss associated with these errors, a loss function $g(\cdot)$ is employed. Common choices for $g(\cdot)$ include the squared error loss, $g(e_t) = e_t^2$, or the absolute error loss, $g(e_t) = |e_t|$. The differential loss at time t is then defined as $d_t = g(e_{1t}) - g(e_{2t})$. The null hypothesis ($H_0$) of the DM test posits that the two forecasts exhibit identical accuracy, implying that the expected value of the loss differential is zero:

$$H_0 : E[d_t] = 0$$

The alternative hypothesis (H_1) asserts that the forecasting accuracies are distinct:

$$H_1 : E[d_t] \neq 0$$

The DM test statistic is constructed as:

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{Var}}(\bar{d})}}$$

where $\bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t$ represents the sample mean of the loss differential over T observations, and $\widehat{\mathrm{Var}}(\bar{d})$ is a consistent estimator of the variance of $\bar{d}$. This variance estimator is crucial, as it accounts for potential serial correlation and heteroskedasticity inherent in the loss differential series d_t. Under the null hypothesis, the DM test statistic asymptotically follows a standard normal distribution, i.e., $DM \sim N(0,1)$. Effective application of the DM test for forecast accuracy assessment necessitates several key prerequisites. First, accurate computation of forecast errors for each model is required, typically defined as the difference between the observed value and the forecasted value. Second, a clearly defined and appropriate loss function must be specified to quantify the cost of forecasting errors in a manner relevant to the specific application, as the selection of the loss function directly impacts the interpretation of "accuracy." Third, as an asymptotic test, the DM test relies on a sufficiently large sample size (T) to ensure that the DM test statistic converges to its asymptotic normal distribution, thereby validating the statistical inference. Finally, while the original formulation assumed covariance stationarity of the loss differential series (d_t), modern extensions of the DM test can accommodate non-stationary series, enhancing its applicability to a broader range of time series data. The p-value derived from the DM test statistic serves as the basis for inference. A p-value below a pre-specified significance level (α, commonly 0.05) leads to the rejection of H_0, indicating a statistically significant difference in forecasting accuracy between the two models.

C. GW TEST

The Giacomini–White (GW) test evaluates conditional predictive ability. Let $\ell_{i,t}$ and $\ell_{j,t}$ denote the CRPS losses of models M_i and M_j at time t, and define the loss differential as

$$d_t = \ell_{i,t} - \ell_{j,t}. \tag{21}$$

The key design choice in the GW framework is the information set on which predictive ability is evaluated. This is encoded through the instrument vector z_t, which is required to be measurable with respect to the forecaster's information available at time t (i.e., before observing d_{t+1}).

$$z_t = (1, d_t), \tag{22}$$

and a one-step alignment ($\tau = 1$). This choice has two motivations. First, the constant instrument 1 ensures that the GW null includes the unconditional equal-loss restriction as a component. Indeed, if $\mathbb{E}[z_t d_{t+1}] = 0$ holds for $z_t \equiv 1$, then $\mathbb{E}[d_{t+1}] = 0$, meaning there is no average CRPS advantage between the two models.

Second, the lagged loss differential d_t is included to test for state dependence in relative performance. Electricity prices and their forecast errors are well known to exhibit strong temporal dependence and regime-like behavior; consequently, it is plausible that the model that "wins" today (lower loss) tends to keep winning tomorrow (persistence), or that relative performance alternates (mean reversion). The instrument d_t provides a minimal way to detect such effects: if d_t contains predictive information about d_{t+1}, then the relative performance of the two models is not conditionally stable over time. In other words, $z_t = (1, d_t)$ tests whether the sign and magnitude of the most recent advantage is informative for the next period.

With this instrument choice, the GW null hypothesis is

$$H_0^{GW} : \mathbb{E}[z_t\, d_{t+1}] = 0, \tag{23}$$

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which jointly imposes (i) $\mathbb{E}[d_{t+1}] = 0$ (no mean CRPS advantage) and (ii) $\mathbb{E}[d_t d_{t+1}] = 0$ (no predictability/persistence in the loss differential from one period to the next).

The GW test yields a nonnegative Wald statistic $GW_{i,j}$ and a p-value computed from a χ^2_q reference distribution with $q = \dim(z_t) = 2$. We report the mean GW statistics in Table 10 and the corresponding p-values in Table 11. Small p-values (below $\alpha = 0.05$) indicate rejection of equal conditional predictive ability. Importantly, because the statistic is quadratic, it is symmetric in the model pair ($GW_{i,j} = GW_{j,i}$); therefore, the GW statistic (and the two-sided GW p-value) quantify whether two models differ, but do not encode which model is better. Direction must be inferred from the sign of $\bar{d}$ (equivalently, from the sign of the DM statistic reported in Table 8).

D. SUPPLEMENTARY GRAPHICS AND TABLES

This appendix provides additional results for Section IV. Figures 11–16 show MPIW, total profit, the number of profitable days, and per-transaction profit plotted against PICP. Tables 4-9 report optimal hyperparameters, total profit, the number of profitable days, per-transaction profit, DM statistics, and DM p-values.

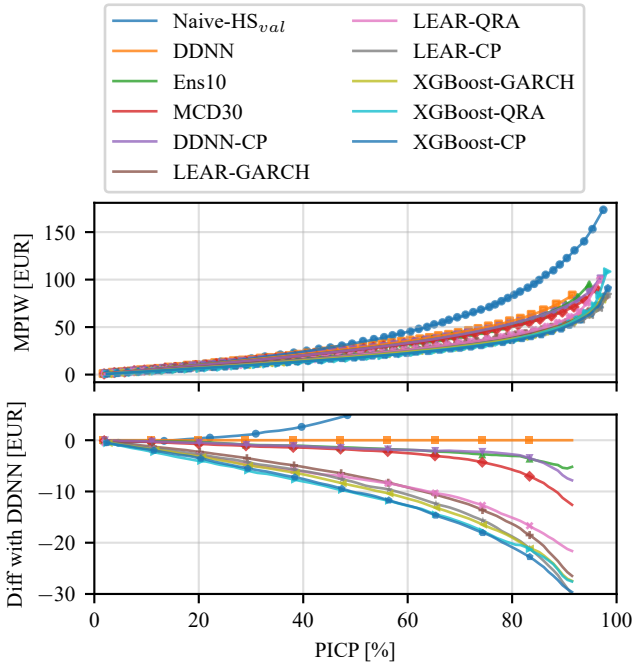

FIGURE 11. Models are evaluated using MPIW across confidence intervals ranging from 2% to 98%, and the results are plotted against PICP. The top subplot shows the average results over 10 independent runs, with standard deviations indicated by dotted lines. The bottom subplot displays the difference in performance between the DDNN and the other models, calculated using piecewise linear interpolation of the curves.

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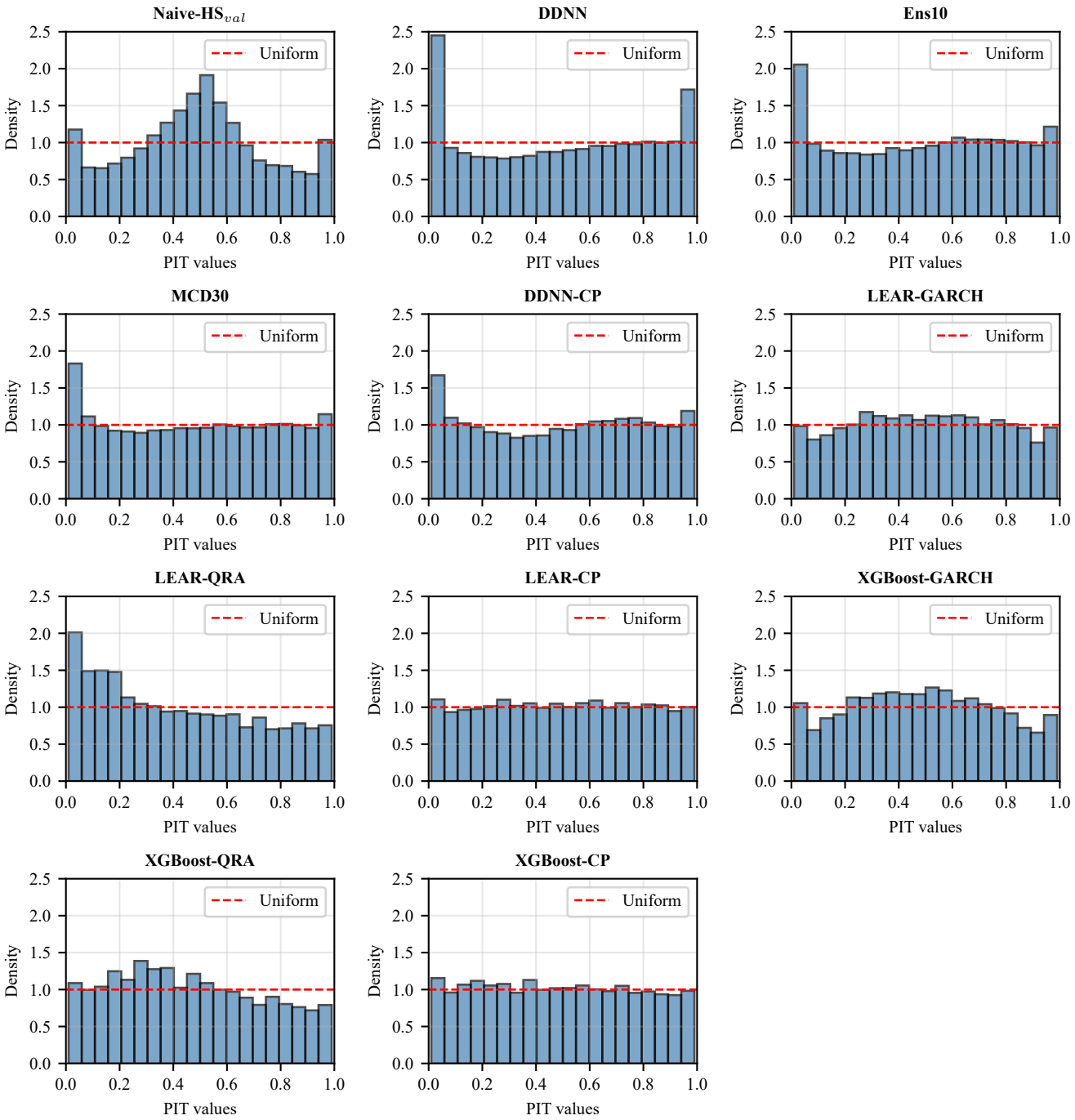

FIGURE 12. **Histograms of the predictive distribution functions, corresponding to each model, evaluated at the actual observations (probability integral transform). The empirical density being close to uniformity indicates a good fit.**

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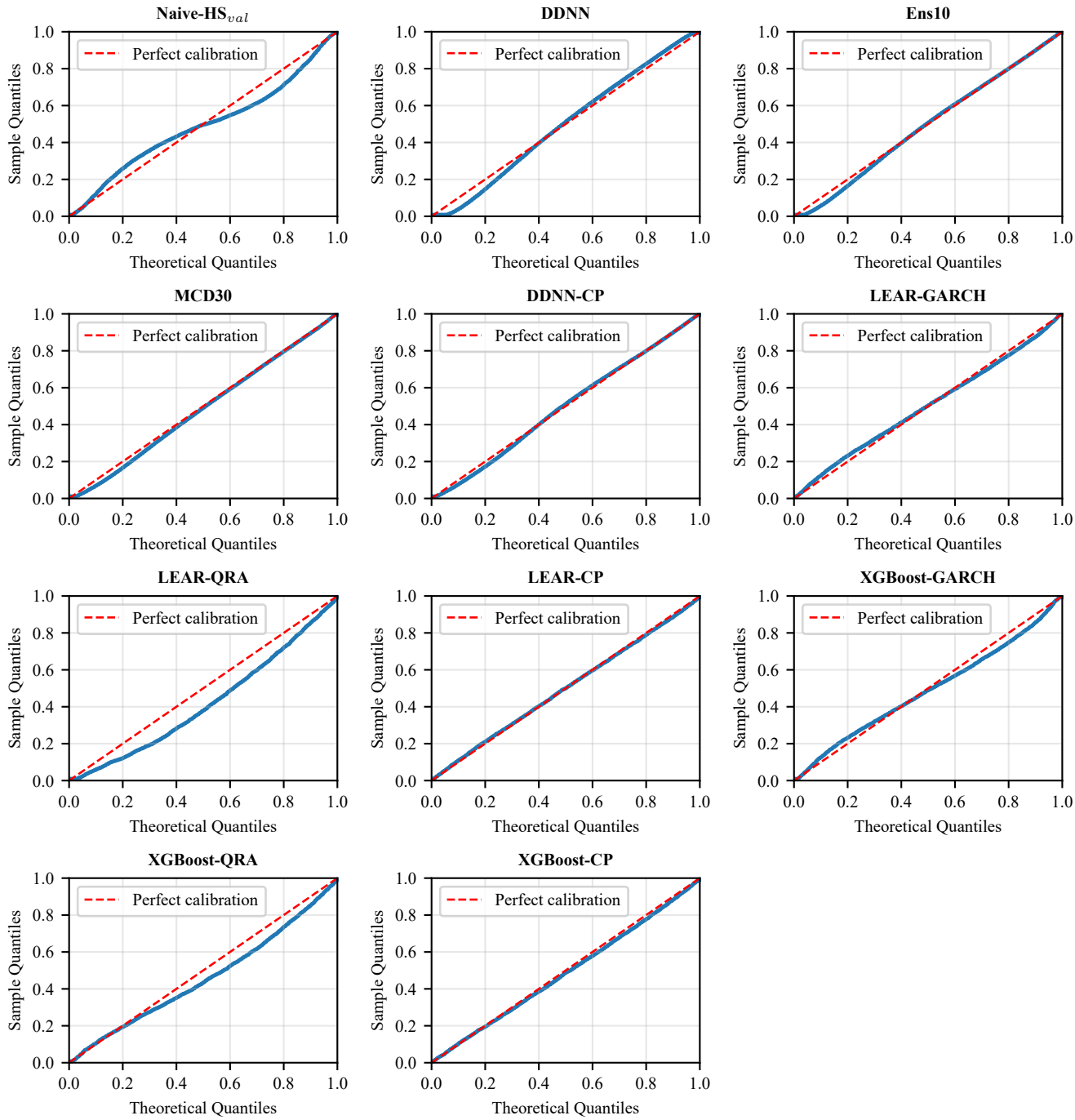

FIGURE 13. TODO

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TABLE 4. Hyperparameters and corresponding search ranges considered during tuning for each model type.

	LEAR(-GARCH/QRA)	DDNN	MCD(10/30)	EvDNN	XGBoost
Learning rate	-	$[10^{-5}, 10^{-1}]$	$[10^{-5}, 10^{-1}]$	$[10^{-5}, 10^{-1}]$	$[10^{-2}, 0.3]$
L_2 regularization parameter	-	$[10^{-5}, 10^{-1}]$	$[10^{-5}, 10^{-1}]$	$[10^{-5}, 10^{-1}]$	$[0.1, 10]$
Dropout rate	-	-	$[0.01, 0.9]$	-	-
Evidence regularization parameter	-	-	-	$[10^{-4}, 1]$	-
LASSO penalty parameter	$[10^{-5}, 10^{-1}]$	-	-	-	-
Number of trees	-	-	-	-	$[50, 500]$
Maximum tree depth	-	-	-	-	$[3, 10]$
Min. child weight	-	-	-	-	$[1, 10]$
Row subsample ratio	-	-	-	-	$[0.6, 1]$
Column subsample ratio	-	-	-	-	$[0.6, 1]$
Min. loss reduction (γ)	-	-	-	-	$[0, 5]$
L_1 regularization parameter (α)	-	-	-	-	$[0, 1]$

TABLE 5. Total profit of models is shown for confidence intervals of 50%, 60%, 70%, 80%, 90% and 98% using the quantile-based strategy. For comparison, the optimal bids strategy is also included. Results are averaged over 10 independent runs, with standard deviations shown in light gray. The best-performing results are highlighted in bold.

Total profit [EUR]	$PI_{50\%}$	$PI_{60\%}$	$PI_{70\%}$	$PI_{80\%}$	$PI_{90\%}$	$PI_{98\%}$
Optimal bids			32543±0			
Naive-HS_{val}	19280 ±0	17876 ±0	15866 ±0	12550 ±0	6669 ±0	1389 ±0
DDNN	25941 ±583	25397 ±900	24120 ±1458	22067 ±1757	17891 ±2592	8491 ±3551
Ens10	26695 ±410	25907 ±496	25014 ±587	22786 ±951	18175 ±1373	5586 ±1864
MCD30	26959 ±645	26754 ±554	26135 ±848	23735 ±1035	18346 ±1644	4217 ±1441
DDNN-CP	24887 ±714	23976 ±873	22281 ±1093	19585 ±1121	14386 ±1304	4699 ±731
Ens10-CP	25679 ±379	24964 ±380	23279 ±546	21065 ±413	15846 ±500	5349 ±322
MCD30-CP	26111 ±534	25462 ±556	24299 ±653	21819 ±957	17212 ±1315	5601 ±755
LEAR-GARCH	27368 ±0	27652 ±0	27749 ±0	26004 ±0	23296 ±0	16485 ±0
LEAR-QRA	**28427** ±0	**28122** ±0	**28476** ±0	**27766** ±0	**25704** ±0	10298 ±0
LEAR-CP	26757 ±0	26653 ±0	26388 ±0	26733 ±0	26224 ±0	**22108** ±0
XGBoost-GARCH	26775 ±0	27665 ±0	26114 ±0	23997 ±0	21295 ±0	14379 ±0
XGBoost-QRA	26823 ±0	27143 ±0	26193 ±0	24913 ±0	20741 ±0	7516 ±0
XGBoost-CP	27154 ±0	27249 ±0	27714 ±0	26775 ±0	24494 ±0	11284 ±0

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TABLE 6. **The expected number of days with profitable limit trades (satisfying Equation 14, 17, or 20) is shown for confidence intervals of 50%, 60%, 70%, 80%, 90% and 98% using the quantile-based strategy. Results are averaged over 10 independent runs, with standard deviations shown in light gray. The best-performing results are highlighted in bold.**

Expected number of days with profitable limit trades [%]	$PI_{50\%}$	$PI_{60\%}$	$PI_{70\%}$	$PI_{80\%}$	$PI_{90\%}$	$PI_{98\%}$
Naive-HS_{val}	65.30 ±0.00	56.28 ±0.00	43.99 ±0.00	31.97 ±0.00	16.44 ±0.00	4.11 ±0.00
DDNN	91.39 ±3.09	84.87 ±4.99	75.35 ±6.51	64.41 ±7.22	47.40 ±7.77	21.01 ±8.08
Ens10	92.50 ±1.13	84.60 ±1.84	75.64 ±2.45	63.01 ±3.18	45.37 ±3.76	12.96 ±4.08
MCD30	94.11 ±1.11	89.24 ±1.35	81.67 ±2.36	69.07 ±4.23	47.21 ±4.79	10.11 ±3.06
DDNN-CP	87.02 ±1.80	78.83 ±2.68	68.55 ±3.47	55.44 ±3.57	37.79 ±3.02	10.79 ±1.72
Ens10-CP	89.34 ±0.59	83.11 ±0.78	71.89 ±1.37	59.78 ±1.11	41.12 ±1.11	12.62 ±0.70
MCD30-CP	91.01 ±0.78	85.05 ±1.03	76.28 ±1.38	63.33 ±2.84	44.86 ±2.87	12.76 ±1.67
LEAR-GARCH	92.90 ±0.00	91.26 ±0.00	84.70 ±0.00	76.23 ±0.00	61.48 ±0.00	37.43 ±0.00
LEAR-QRA	95.36 ±0.00	91.80 ±0.00	88.52 ±0.00	81.97 ±0.00	69.67 ±0.00	24.86 ±0.00
LEAR-CP	92.90 ±0.00	91.53 ±0.00	**89.62** ±0.00	**86.89** ±0.00	**82.51** ±0.00	**60.66** ±0.00
XGBoost-GARCH	90.44 ±0.00	87.43 ±0.00	78.69 ±0.00	69.40 ±0.00	56.01 ±0.00	31.15 ±0.00
XGBoost-QRA	91.80 ±0.00	87.16 ±0.00	78.14 ±0.00	66.39 ±0.00	47.27 ±0.00	13.66 ±0.00
XGBoost-CP	**95.63** ±0.00	**93.44** ±0.00	88.52 ±0.00	77.87 ±0.00	64.21 ±0.00	20.49 ±0.00

TABLE 7. **Per-transaction profit of the models is shown for confidence intervals of 50%, 60%, 70%, 80%, 90% and 98% using the quantile-based strategy. For comparison, the optimal bids strategy is also included. Results are averaged over 10 independent runs, with standard deviations shown in light gray. The best-performing results are highlighted in bold.**

Per-transaction profit [EUR]	$PI_{50\%}$	$PI_{60\%}$	$PI_{70\%}$	$PI_{80\%}$	$PI_{90\%}$	$PI_{98\%}$
Optimal bids	44.46±0.00					
Naive-HS_{val}	**44.43** ±0.00	**46.55** ±0.00	**50.21** ±0.00	**54.56** ±0.00	**59.54** ±0.00	46.29 ±0.00
DDNN	40.80 ±1.12	42.70 ±1.48	45.50 ±1.77	48.45 ±1.80	52.92 ±1.91	55.04 ±3.58
Ens10	41.21 ±0.45	43.42 ±0.58	46.32 ±0.70	50.18 ±1.17	55.18 ±0.68	58.71 ±2.31
MCD30	40.81 ±1.04	42.14 ±1.10	44.58 ±1.29	47.43 ±1.30	53.62 ±1.14	56.35 ±4.04
DDNN-CP	41.88 ±0.99	43.83 ±1.02	46.22 ±0.99	49.78 ±0.97	52.82 ±1.75	60.29 ±3.70
Ens10-CP	41.65 ±0.43	42.73 ±0.57	45.79 ±0.65	49.17 ±0.68	53.25 ±0.98	58.02 ±1.00
MCD30-CP	41.31 ±0.98	42.48 ±1.21	44.73 ±0.86	47.88 ±0.75	52.90 ±1.10	60.45 ±2.63
LEAR-GARCH	43.44 ±0.00	43.61 ±0.00	45.79 ±0.00	46.94 ±0.00	51.77 ±0.00	60.16 ±0.00
LEAR-QRA	42.94 ±0.00	43.94 ±0.00	45.20 ±0.00	47.54 ±0.00	50.80 ±0.00	56.58 ±0.00
LEAR-CP	42.07 ±0.00	41.78 ±0.00	42.02 ±0.00	42.98 ±0.00	44.00 ±0.00	50.02 ±0.00
XGBoost-GARCH	43.19 ±0.00	44.48 ±0.00	45.34 ±0.00	47.24 ±0.00	51.94 ±0.00	63.07 ±0.00
XGBoost-QRA	41.27 ±0.00	43.78 ±0.00	46.28 ±0.00	51.47 ±0.00	59.95 ±0.00	**75.16** ±0.00
XGBoost-CP	42.30 ±0.00	41.79 ±0.00	43.30 ±0.00	47.14 ±0.00	52.34 ±0.00	75.23 ±0.00

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TABLE 8. Pairwise DM statistics $\widehat{DM}_{i,j}$ (row model i vs. column model j). Negative values favor the row model, positive values favor the column model. Results are averaged over 10 independent runs, yielding 100 comparisons per model pair across runs.

	Naive-HS$_{val}$	DDNN	Ens10	MCD30	LEAR-GARCH	LEAR-QRA	DDNN-CP	LEAR-CP	Ens10-CP	MCD30-CP
Naive-HS$_{val}$	–	6.273 ±0.660	6.984 ±0.157	7.387 ±0.356	10.619 ±0.000	10.514 ±0.000	6.433 ±0.638	10.235 ±0.000	6.898 ±0.193	7.232 ±0.319
DDNN	-6.273 ±0.660	–	2.432 ±2.309	3.087 ±1.868	5.763 ±0.712	5.781 ±0.692	0.495 ±2.370	5.400 ±0.703	2.037 ±2.179	2.650 ±1.801
Ens10	-6.984 ±0.157	-2.432 ±2.309	–	1.748 ±1.396	5.150 ±0.231	5.184 ±0.240	-1.778 ±2.084	4.784 ±0.225	-0.299 ±1.419	1.201 ±1.244
MCD30	-7.387 ±0.356	-3.087 ±1.868	-1.748 ±1.396	–	4.683 ±0.361	4.692 ±0.388	-2.700 ±1.782	4.313 ±0.358	-1.856 ±1.312	-0.438 ±1.555
LEAR-GARCH	-10.619 ±0.000	-5.763 ±0.712	-5.150 ±0.231	-4.683 ±0.361	–	-0.072 ±0.000	-5.731 ±0.658	-0.049 ±0.000	-5.264 ±0.219	-4.815 ±0.340
LEAR-QRA	-10.514 ±0.000	-5.781 ±0.692	-5.184 ±0.240	-4.692 ±0.388	0.072 ±0.000	–	-5.752 ±0.636	0.039 ±0.000	-5.305 ±0.227	-4.839 ±0.377
DDNN-CP	-6.433 ±0.638	-0.495 ±2.370	1.778 ±2.084	2.700 ±1.782	5.731 ±0.658	5.752 ±0.636	–	5.374 ±0.658	1.656 ±2.251	2.431 ±1.816
LEAR-CP	-10.235 ±0.000	-5.400 ±0.703	-4.784 ±0.225	-4.313 ±0.358	0.049 ±0.000	-0.039 ±0.000	-5.374 ±0.658	–	-4.921 ±0.224	-4.474 ±0.340
Ens10-CP	-6.898 ±0.193	-2.037 ±2.179	0.299 ±1.419	1.856 ±1.312	5.264 ±0.219	5.305 ±0.227	-1.656 ±2.251	4.921 ±0.224	–	1.528 ±1.336
MCD30-CP	-7.232 ±0.319	-2.650 ±1.801	-1.201 ±1.244	0.438 ±1.555	4.815 ±0.340	4.839 ±0.377	-2.431 ±1.816	4.474 ±0.340	-1.528 ±1.336	–

TABLE 9. Pairwise DM p-values $p_{i,j}$ (row model i vs. column model j). Results are averaged over 10 independent runs, yielding 100 comparisons per model pair across runs. For notational convenience, we write $p = 0.000$ whenever $p < 0.001$.

	Naive-HS$_{val}$	DDNN	Ens10	MCD30	LEAR-GARCH	LEAR-QRA	DDNN-CP	LEAR-CP	Ens10-CP	MCD30-CP
Naive-HS$_{val}$	–	1.000 ±0.000	1.000 ±0.000	1.000 ±0.000	1.000 ±0.000	1.000 ±0.000	1.000 ±0.000	1.000 ±0.000	1.000 ±0.000	1.000 ±0.000
DDNN	0.000 ±0.000	–	0.808 ±0.299	0.930 ±0.181	1.000 ±0.000	1.000 ±0.000	0.578 ±0.413	1.000 ±0.000	0.776 ±0.312	0.903 ±0.194
Ens10	0.000 ±0.000	0.192 ±0.299	–	0.870 ±0.278	1.000 ±0.000	1.000 ±0.000	0.243 ±0.324	1.000 ±0.000	0.422 ±0.349	0.809 ±0.277
MCD30	0.000 ±0.000	0.070 ±0.181	0.130 ±0.278	–	1.000 ±0.000	1.000 ±0.000	0.090 ±0.203	1.000 ±0.000	0.118 ±0.268	0.376 ±0.289
LEAR-GARCH	0.000 ±0.000	0.000 ±0.000	0.000 ±0.000	0.000 ±0.000	–	0.471 ±0.000	0.000 ±0.000	0.480 ±0.000	0.000 ±0.000	0.000 ±0.000
LEAR-QRA	0.000 ±0.000	0.000 ±0.000	0.000 ±0.000	0.000 ±0.000	0.529 ±0.000	–	0.000 ±0.000	0.516 ±0.000	0.000 ±0.000	0.000 ±0.000
DDNN-CP	0.000 ±0.000	0.422 ±0.413	0.757 ±0.324	0.910 ±0.203	1.000 ±0.000	1.000 ±0.000	–	1.000 ±0.000	0.722 ±0.348	0.882 ±0.220
LEAR-CP	0.000 ±0.000	0.000 ±0.000	0.000 ±0.000	0.000 ±0.000	0.520 ±0.000	0.484 ±0.000	0.000 ±0.000	–	0.000 ±0.000	0.000 ±0.000
Ens10-CP	0.000 ±0.000	0.224 ±0.312	0.578 ±0.349	0.882 ±0.268	1.000 ±0.000	1.000 ±0.000	0.278 ±0.348	1.000 ±0.000	–	0.846 ±0.274
MCD30-CP	0.000 ±0.000	0.097 ±0.194	0.191 ±0.277	0.624 ±0.289	1.000 ±0.000	1.000 ±0.000	0.118 ±0.220	1.000 ±0.000	0.154 ±0.274	–

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TABLE 10. Pairwise GW statistics $\widehat{GW}_{i,j}$ (row model i vs. column model j). Entries report the mean (black) and standard deviation (grey) of the GW Wald statistic across 10 independent runs, yielding 100 comparisons per model pair across runs. The GW statistic is nonnegative and symmetric; larger values indicate stronger evidence against equal conditional predictive ability (with instruments $z_t = (1, d_t)$), but the statistic itself does not encode directional superiority.

	Naive-HS$_{val}$	DDNN	Ens10	MCD30	LEAR-GARCH	LEAR-QRA	DDNN-CP	LEAR-CP	Ens10-CP	MCD30-CP	XGBoost-GARCH	XGBoost-QRA	XGBoost-CP
Naive-HS$_{val}$	–	207.516 ±31.567	241.693 ±10.253	268.074 ±23.909	569.475 ±0.000	554.809 ±0.000	220.843 ±29.438	518.754 ±0.000	241.890 ±10.917	261.621 ±19.119	710.534 ±0.000	662.433 ±0.000	631.553 ±0.000
DDNN	207.516 ±31.567	–	192.107 ±66.369	225.258 ±64.784	227.447 ±28.958	235.151 ±32.162	148.039 ±83.445	218.213 ±32.990	154.522 ±77.985	190.716 ±69.191	234.522 ±28.563	276.635 ±35.413	231.228 ±27.153
Ens10	241.693 ±10.253	192.107 ±66.369	–	250.604 ±51.804	197.816 ±9.144	211.024 ±9.661	154.317 ±59.896	191.470 ±10.220	86.728 ±55.405	185.482 ±64.800	190.199 ±10.636	231.857 ±13.530	189.173 ±9.938
MCD30	268.074 ±23.909	225.258 ±64.784	250.604 ±51.804	–	176.312 ±22.452	182.922 ±21.929	204.113 ±68.979	170.700 ±17.715	214.412 ±79.587	163.062 ±93.949	165.207 ±24.737	197.001 ±31.791	163.750 ±25.865
LEAR-GARCH	569.475 ±0.000	227.447 ±28.958	197.816 ±9.144	176.312 ±22.452	–	151.411 ±0.000	209.344 ±28.459	6.653 ±0.000	166.855 ±10.649	145.468 ±18.520	51.421 ±0.000	70.030 ±0.000	85.964 ±0.000
LEAR-QRA	554.809 ±0.000	235.151 ±32.162	211.024 ±9.661	182.922 ±21.929	151.411 ±0.000	–	215.372 ±31.311	62.650 ±0.000	175.030 ±12.450	148.262 ±21.748	39.524 ±0.000	42.912 ±0.000	56.547 ±0.000
DDNN-CP	220.843 ±29.438	148.039 ±83.445	154.317 ±59.896	204.113 ±68.979	209.344 ±28.459	215.372 ±31.311	–	206.203 ±29.832	159.858 ±72.268	202.897 ±69.852	218.218 ±32.474	267.833 ±38.570	208.056 ±28.027
LEAR-CP	518.754 ±0.000	218.213 ±32.990	191.470 ±10.220	170.700 ±17.715	6.653 ±0.000	62.650 ±0.000	206.203 ±29.832	–	172.392 ±10.611	148.807 ±18.363	14.816 ±0.000	19.340 ±0.000	19.300 ±0.000
Ens10-CP	241.890 ±10.917	154.522 ±77.985	86.728 ±55.405	214.412 ±79.587	166.855 ±10.649	175.030 ±12.450	159.858 ±72.268	172.392 ±10.611	–	229.840 ±51.211	173.916 ±12.551	218.942 ±16.587	164.802 ±11.199
MCD30-CP	261.621 ±19.119	190.716 ±69.191	185.482 ±64.800	163.062 ±93.949	145.468 ±18.520	148.262 ±21.748	202.897 ±69.852	148.807 ±18.363	229.840 ±51.211	–	147.177 ±17.228	178.324 ±22.419	140.200 ±16.801
XGBoost-GARCH	710.534 ±0.000	234.522 ±28.563	190.199 ±10.636	165.207 ±24.737	51.421 ±0.000	39.524 ±0.000	218.218 ±32.474	14.816 ±0.000	173.916 ±12.551	147.177 ±17.228	–	46.221 ±0.000	12.435 ±0.000
XGBoost-QRA	662.433 ±0.000	276.635 ±35.413	231.857 ±13.530	197.001 ±31.791	70.030 ±0.000	42.912 ±0.000	267.833 ±38.570	19.340 ±0.000	218.942 ±16.587	178.324 ±22.419	46.221 ±0.000	–	34.185 ±0.000
XGBoost-CP	631.553 ±0.000	231.228 ±27.153	189.173 ±9.938	163.750 ±25.865	85.964 ±0.000	56.547 ±0.000	208.056 ±28.027	19.300 ±0.000	164.802 ±11.199	140.200 ±16.801	12.435 ±0.000	34.185 ±0.000	–

TABLE 11. Pairwise GW p-values $\hat{p}^{GW}_{i,j}$ (row model i vs. column model j). Entries report the mean (black) and standard deviation (grey) across 10 independent runs, yielding 100 comparisons per model pair across runs. Values below $\alpha = 0.05$ indicate statistically significant differences in conditional predictive ability in the direction favoring the row model.

	Naive-HS$_{val}$	DDNN	Ens10	MCD30	LEAR-GARCH	LEAR-QRA	DDNN-CP	LEAR-CP	Ens10-CP	MCD30-CP	XGBoost-GARCH	XGBoost-QRA	XGBoost-CP
Naive-HS$_{val}$	–	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0
DDNN	1.20e-36 ±3.58e-36	–	0.82 ±0.38	0.95 ±0.22	1.00 ±0	1.00 ±0	0.59 ±0.49	1.00 ±0	0.78 ±0.41	0.95 ±0.22	1.00 ±0	1.00 ±0	1.00 ±0
Ens10	7.40e-51 ±1.83e-50	0.18 ±0.38	–	0.90 ±0.30	1.00 ±0	1.00 ±0	0.24 ±0.43	1.00 ±0	0.39 ±0.49	0.90 ±0.30	1.00 ±0	1.00 ±0	1.00 ±0
MCD30	4.57e-46 ±1.37e-45	0.05 ±0.22	0.10 ±0.30	–	1.00 ±0	1.00 ±0	0.06 ±0.24	1.00 ±0	0.10 ±0.30	0.33 ±0.47	1.00 ±0	1.00 ±0	1.00 ±0
LEAR-GARCH	2.19e-124 ±4.2e-14	7.61e-44 ±2.05e-43	6.09e-41 ±1.60e-40	1.30e-33 ±3.61e-33	–	1.32e-33 ±0	1.46e-36 ±4.38e-36	0.04 ±1.39e-17	1.97e-34 ±4.66e-34	3.60e-27 ±7.21e-27	1.00 ±0	1.00 ±0	1.00 ±0
LEAR-QRA	3.35e-121 ±4.30e-137	9.20e-43 ±2.76e-42	9.19e-44 ±2.46e-43	3.67e-35 ±9.82e-35	1.00 ±0	–	3.42e-38 ±1.02e-37	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0	1.00 ±0
DDNN-CP	5.63e-39 ±1.69e-38	0.41 ±0.49	0.76 ±0.43	0.94 ±0.24	1.00 ±0	1.00 ±0	–	1.00 ±0	0.71 ±0.45	0.93 ±0.26	1.00 ±0	1.00 ±0	1.00 ±0
LEAR-CP	2.26e-113 ±8.66e-129	8.47e-42 ±1.85e-41	1.70e-39 ±3.77e-39	6.17e-34 ±1.36e-33	1.00 ±0	2.49e-14 ±1.26e-29	8.01e-38 ±2.40e-37	–	1.04e-35 ±2.61e-35	6.74e-28 ±1.43e-27	1.00 ±0	1.00 ±2.71e-20	1.00 ±0
Ens10-CP	2.54e-50 ±5.73e-50	0.22 ±0.41	0.61 ±0.49	0.90 ±0.30	1.00 ±0	1.00 ±0	0.29 ±0.45	1.00 ±0	–	0.90 ±0.30	1.00 ±0	1.00 ±0	1.00 ±0
MCD30-CP	9.67e-48 ±2.90e-47	0.05 ±0.22	0.10 ±0.30	0.67 ±0.47	1.00 ±0	1.00 ±0	0.07 ±0.26	1.00 ±0	0.10 ±0.30	–	1.00 ±0	1.00 ±0	1.00 ±0
XGBoost-GARCH	5.12e-155 ±0	6.76e-44 ±1.99e-43	4.01e-39 ±9.58e-39	4.39e-32 ±8.40e-32	6.83e-12 ±0	2.61e-09 ±4.14e-25	1.00e-37 ±3.00e-37	6.06e-04 ±0	7.93e-36 ±1.82e-35	1.09e-29 ±2.79e-29	–	9.19e-11 ±1.29e-26	1.99e-03 ±4.34e-19
XGBoost-QRA	1.43e-144 ±2.85e-16	9.31e-52 ±2.56e-51	1.56e-47 ±3.13e-47	1.03e-37 ±1.93e-37	6.21e-16 ±1.97e-31	4.81e-10 ±0	3.84e-47 ±1.15e-46	6.30e-05 ±2.71e-20	7.17e-45 ±1.40e-44	3.30e-35 ±9.21e-35	1.00 ±0	–	1.00 ±0
XGBoost-CP	7.25e-138 ±4.77e-153	3.07e-44 ±9.10e-44	6.39e-39 ±1.72e-38	5.10e-31 ±9.95e-31	2.15e-19 ±4.81e-35	5.26e-13 ±2.02e-28	3.08e-36 ±9.25e-36	6.40e-05 ±0	5.68e-34 ±1.39e-33	9.75e-28 ±2.65e-27	1.00 ±0	3.77e-08 ±6.62e-24	–

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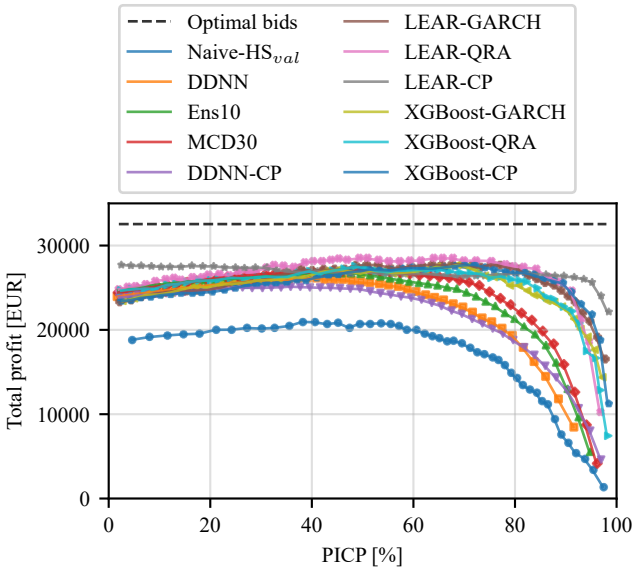

FIGURE 14. Total profit of models is shown for confidence intervals ranging from 2% to 98% using the quantile-based strategy, and the results are plotted against PICP. For comparison, the optimal bids strategy is also included. Results are averaged over 10 independent runs.

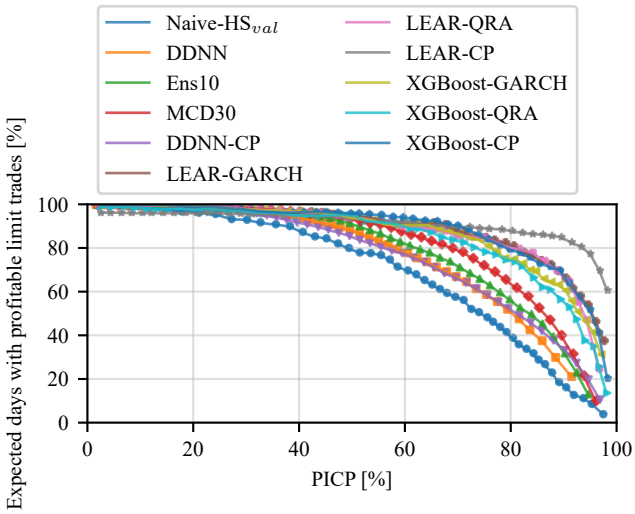

FIGURE 15. The expected number of days with profitable limit trades (satisfying Equation 14, 17, or 20) is shown for each model across confidence intervals ranging from 2% to 98% using the quantile-based strategy and the results are plotted against PICP. Results are averaged over 10 independent runs.

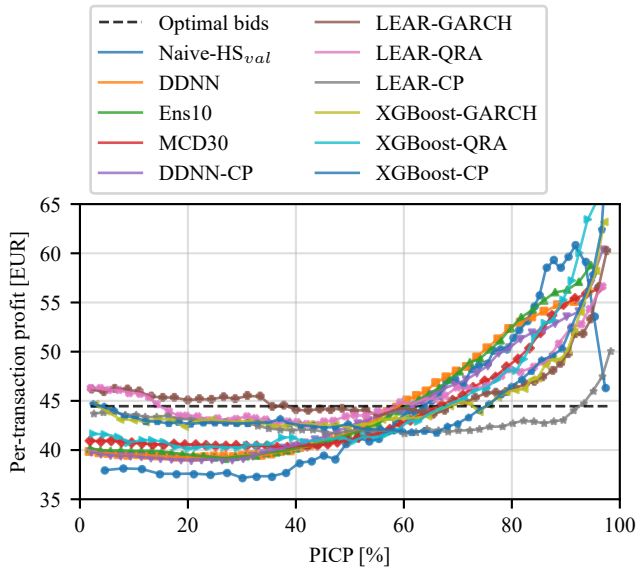

FIGURE 16. Per-transaction profit of models is shown for confidence intervals ranging from 2% to 98% using the quantile-based strategy, and the results are plotted against PICP. For comparison, the optimal bids strategy is also included. Results are averaged over 10 independent runs.

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